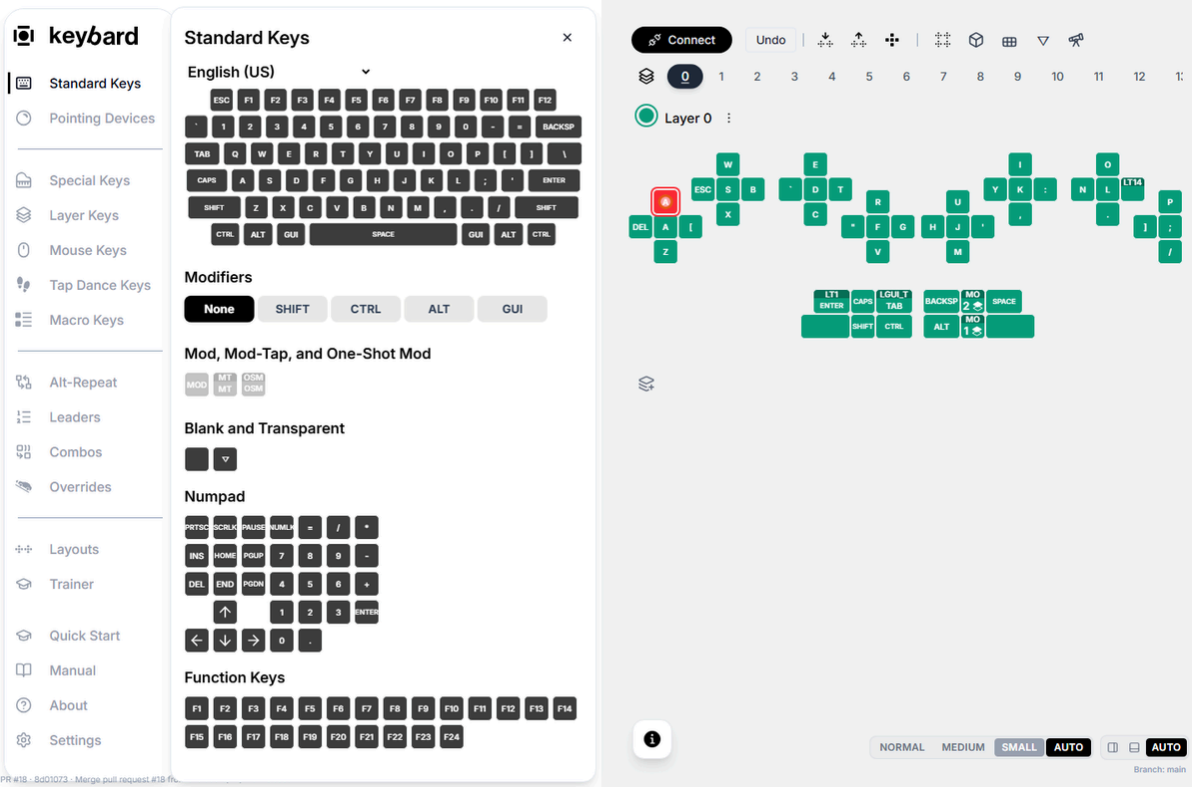


The Keyboard user manual.

Connect your board, make your first change, and learn every part of Keyboard.

11 chapters Illustrated walkthroughs Browser editor + desktop Trainer



The screenshot displays the Keyboard software interface. On the left is a sidebar with a menu containing: keyboard, Standard Keys, Pointing Devices, Special Keys, Layer Keys, Mouse Keys, Tap Dance Keys, Macro Keys, Alt-Repeat, Leaders, Combos, Overrides, Layouts, Trainer, Quick Start, Manual, About, and Settings. The main area is divided into two panels. The left panel, titled 'Standard Keys', shows a keyboard layout for 'English (US)' with various modifier options (None, SHIFT, CTRL, ALT, GUI) and sections for 'Mod, Mod-Tap, and One-Shot Mod', 'Blank and Transparent', 'Numpad', and 'Function Keys'. The right panel shows a keyboard layout editor with a 'Connect' button, an 'Undo' button, and a grid of keys. A key labeled 'A' is highlighted in red, indicating it has been dragged onto the 'Q' key. At the bottom right, there are buttons for 'NORMAL', 'MEDIUM', 'SMALL', and 'AUTO', along with a 'Branch: main' indicator.

Open the panel. Drag a letter. See the change. Open Standard Keys, drag A onto Q, and release. The key now shows A. Actual offline example; no keyboard writes.

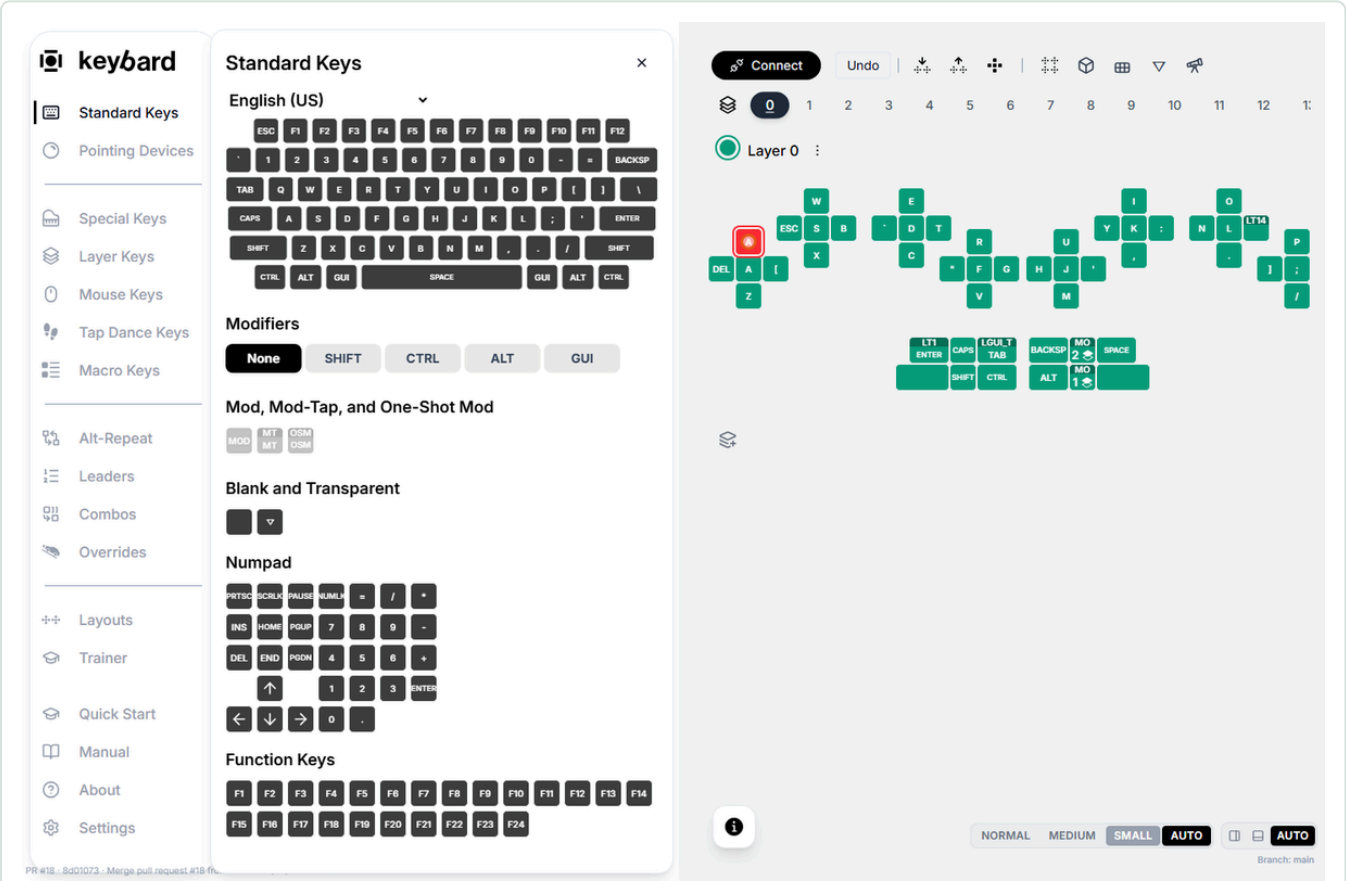
Written against Keyboard `8d01073`. Screenshots use the bundled example and may show fewer behavior slots than current firmware. Connected hardware exposes additional controls. All screenshots can be enlarged.

Start with a keyboard—or an example

Keyboard lets you configure your Svalboard, organize layers, create advanced bindings, and set up an on-screen trainer. You can explore the editor before connecting hardware. This guide starts with one ordinary key change, then builds toward reusable layouts and reliable backups.

Choose a starting point on the welcome screen:

Choose	What happens	Use it for
Connect Keyboard	The browser asks which keyboard to open, then Keyboard reads its configuration.	Editing your connected Svalboard.
Load File	A saved layout opens as an offline draft.	Inspecting a backup or continuing exported work.
QWERTY Example	A bundled example opens without a keyboard.	Learning the controls safely.



Open an example and make a first edit. Open QWERTY Example, open Standard Keys, and drag A directly onto Q. This is an offline draft; no keyboard is changed. Offline example.

Three places your work can live

ON YOUR SVALBOARD

The saved configuration

Runs on the keyboard.
Connecting reads it;
applying edits changes it.

IN THIS BROWSER SESSION

The current draft

May include unsaved
changes. The layout you
see is not always what the
board runs.

ON YOUR COMPUTER

An exported file

A copy you keep. Export
downloads it; import
reviews which parts can be
applied.

Take the five-minute tour

1. Choose **QWERTY Example**.
2. If only icons are showing, click the logo at the top of the left navigation rail to reveal its labels. Hovering an icon also identifies it.
3. Open **Standard Keys** and drag a letter from the palette onto the position you want to change. Release it and check the new legend.
4. Switch layer tabs to see how the same physical positions can carry different bindings.
5. Choose **Export Layout** in the editor toolbar if you want to keep your experiment.

Offline means no keyboard writes. An example or loaded file is a draft in the current session. Export it before closing or refreshing. Connecting a keyboard later reads that keyboard; it does not automatically install your draft.

Demonstrations play automatically when visible. Autoplay is enabled even when your system requests reduced motion.

Most illustrations use the bundled offline example. Captions explicitly identify connected controls shown with controlled test-board data. No screenshot is presented as proof of a hardware write.

CHAPTER 02 / CONNECT YOUR BOARD

Connect and protect your starting point

[Welcome](#) → [Connect Keyboard](#) → [browser device chooser](#)

Use a desktop browser that supports WebHID, such as Chrome or Edge, and a USB data connection. The connection button is unavailable when the browser lacks the required API. Keyboard's browser editor expects compatible Svalboard firmware; seeing a keyboard in a device chooser does not establish firmware compatibility.

1. Connect your Svalboard by USB, open Keyboard, and choose **Connect Keyboard**.
2. Select the intended keyboard in the browser's chooser and grant access. With several keyboards attached, check which one you selected.
3. Wait for the layout to load. Hover the connection/update controls to check the editing target before changing anything.
4. Choose **Export Layout**. Keep the native **.svil (Recommended)** and **Include Macros** enabled, then choose **Export**.
5. Name the downloaded file clearly—for example, `before-first-edit.svil`—and keep it somewhere you can find again.

Back up before editing. Export captures the configuration currently shown in the editor, including staged changes. Exporting after experiments does not recover the untouched configuration you originally connected to.

 keyboard

 Standard Keys

 Pointing Devices

 Special Keys

 Layer Keys

 Mouse Keys

 Tap Dance Keys

 Macro Keys

 Alt-Repeat

 Leaders

 Combos

 Overrides

 Layouts

 Trainer

 Quick Start

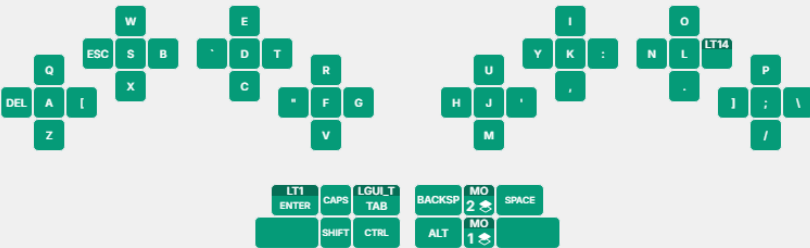
 Manual

 About

Connect

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 mouse

Layer 0





NORMAL MEDIUM SMALL **AUTO**

  **AUTO**

Branch: main

PR #18 - 8d01073 - Merge pull request #18 from svalboard/fix/host-release-link

Export a backup. Open Export Layout and export the native .svil file. The recording verifies a downloaded file; this example uses the browser download fallback. Browsers with a native save picker show a Save dialog instead. Offline example.

Coming from an older configuration tool

Keep your existing layout backup before changing firmware. You can open a legacy .v1l file offline to inspect it, but Keyboard blocks direct import of old-protocol layouts into connected Sval firmware when keycode migration is required. Renaming the file extension is not a conversion.

For Svalboard Vial-QMK v2025-11-01 with the vial keymap, the firmware can migrate supported configuration on first boot. Follow the [firmware upgrade instructions](#), connect in Keyboard, and verify layers, macros, modifier options, timing and pointing settings. Then export a fresh native .svil . Other old or custom builds are not covered; keep the old file as a reference and recreate anything not migrated. A supported firmware migration is different from importing a legacy file in the browser.

Understand the editing target

State	Meaning
Editing keyboard	Changes can be written to the connected device.
Offline draft	Changes affect the loaded file or example in this session. Export to keep them.
Pending changes	The editor has changes waiting to be written; the displayed layout may differ from the board.
Not saved	A write failed. Some writes may already have succeeded; follow the retry guidance.

CHAPTER 03 / YOUR FIRST EDIT

Make and verify your first change

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layers

Trainer

Quick Start

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Settings

Connect

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 mouse

Layer 0

W

ESC S B

Q DEL A I Z

X

E

· D T

C

R

" F G

V

I

U Y K :

H J ' ,

M

O

N L LT14

P

] ; \

/

LT1 ENTER CAPS LGU LT TAB

SHIFT CTRL

BACKSP MO 2

ALT MO 1 SPACE

NORMAL MEDIUM SMALL AUTO

Branch: main

An offline example of the editing workspace: navigation on the left, layer selection above the layout.

Start with one easy-to-recognize binding on a known layer. Keep the keys you need for navigation and switching layers intact until you are comfortable restoring changes.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layouts

Trainer

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Settings

QMK Settings

QMK Settings

Configure firmware behavior. Changes follow your update mode.

Magic

Grave Escape

Tap-Hold

Tapping Term

Permissive Hold (new)

Hold On Other Key Press

Retro Tapping (new)

Quick Tap Term

Tap Code Delay

Tap Hold Caps Delay

Tapping Toggle

Chordal Hold

Flow Tap

Auto Shift

Combo

One Shot Keys

Mouse keys

No pending change

0 1 2 3 4 5 6 7 8 9 10 11 12 13

Layer 0

Q W E I O P

DEL A S B C D T R U Y K L N

Z X C F G H J M

LT1 ENTER CAPS LGU/LT TAB BACKSP MO 2 SPACE

SHIFT CTRL ALT MO 1

NORMAL MEDIUM SMALL AUTO

AUTO

Review and apply a queued change. Change one setting in Manual Updates, check that Pending shows 1, then choose Apply 1 Change. The write endpoint is simulated test-board data; no physical keyboard is changed.

Choose when changes reach the board

Live Updating is on by default. While connected, ordinary queued edits are saved as you make them. For a controlled first experiment, use the lightning control labeled **Switch to Manual Updates**. You can then review pending edits before applying them.

Control	Effect
Live Updating	Edits are saved as you make them; watch for saving or error messages.
Apply N Changes	Writes the pending changes in Manual mode.
Discard pending edits	Discards the current unsaved edits. It does not reverse changes already applied.
Switch to Live Updating	Applies pending changes before enabling Live Updating.

1. Select the layer you intend to edit.
2. Open **Standard Keys**.
3. Drag the replacement key from the palette onto the target position and release.
Alternatively, select the position first and click a palette key.

4. Check the resulting legend. In connected Manual mode, **Pending (N)** counts the edits waiting to be written.
5. Choose **Apply N Changes** and wait for saving to finish. Do not assume a changed legend alone means the board was updated.
6. Deselect the key, switch to a text editor, activate the appropriate layer, and test the result.
7. Export the verified configuration as a new file.

The screenshot displays the Keyboard software interface. On the left is a sidebar with navigation options: Standard Keys, Pointing Devices, Special Keys, Layer Keys, Mouse Keys, Tap Dance Keys, Macro Keys, Alt-Repeat, Leaders, Combos, Overrides, Layouts, Trainer, Quick Start, Manual, and About. The main area is divided into two panels. The left panel, titled 'Standard Keys', shows a standard QWERTY keyboard layout for English (US). Below it are sections for 'Modifiers' (None, SHIFT, CTRL, ALT, GUI), 'Mod, Mod-Tap, and One-Shot Mod' (MOD, MT, OSM), 'Blank and Transparent' (a dropdown menu), 'Numpad' (a numeric keypad layout), and 'Function Keys' (a function key layout). The right panel shows a custom keyboard layout for 'Layer 0'. It features a compact design with keys like W, E, D, T, R, U, C, F, G, H, J, K, M, V, Z, A, S, X, B, N, and M. At the bottom of the right panel, there are settings for 'NORMAL', 'MEDIUM', 'SMALL', and 'AUTO' (selected), along with a 'Branch: main' indicator.

Drag A onto the Q position. Watch the pointer pick up A from Standard Keys, drag it onto Q, and release. The position now shows A. This changes the offline example only.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layers

Trainer

Quick Start

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Connect

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 mouse

Layer 0

Q

ESC

W

S

B

DEL

Q

[

Z

X

E

D

T

C

R

F

G

V

U

H

J

M

Y

K

:

I

N

L

LT14

P

;

\

/

LT1

ENTER

CAPS

LOUT

TAB

BACKSP

MO

2

SPACE

SHIFT

CTRL

ALT

MO

1

PR #18 - B401073 - Merge pull request #18 from svalboard/fix/host-release-link

Branch: main

Swap two existing keys. Drag Q onto A to exchange their bindings, then drag them back to restore the original layout. Palette-to-layout dragging assigns; layout-to-layout dragging swaps. Offline example.

Typing can assign a key. Typing Binds a Key is enabled by default. With an editing target selected, pressing a physical key can assign that key, including modifiers, to the target. Deselect before testing normal typing, or turn this option off in **Settings** → **General**.

Dragging an existing layout key onto another position swaps bindings, including across layers. Use this deliberately: it is different from copying a palette key. If an **Undo** button appears, its tooltip identifies the operation it can reverse; do not treat it as unlimited editing history.

CHAPTER 04 / WORK WITH LAYERS

Build a layout you can navigate

A layer is another set of bindings for the same physical key positions. Editing a layer tab changes what you are looking at in Keyboard; using a layer binding on the physical keyboard changes what the keyboard types.

Term	Meaning
Base/default layer	The starting layer used when no higher active layer provides a binding.
Transparent key	Looks through to a lower active layer for its binding.
Blank key	Produces no action at that position instead of falling through.
Momentary layer key	Activates another layer while held.
Layer-tap key	Types one action when tapped and activates a layer when held.

Open **Layer Keys** for layer bindings. Start with a momentary key leading to a small navigation or symbols layer. Keep the activation key usable while that layer is active; a transparent binding at the corresponding position is often appropriate.

The screenshot displays the Keyboard Maestro interface. On the left, the 'Layer Keys' window is open, showing a list of layers (0-15) with their respective momentary keys. Layer 3 is highlighted. The main window shows a keyboard layout diagram with keys assigned to various layers. The 'Layer 0' diagram shows a standard QWERTY layout with additional function keys. The 'Layer 3' diagram shows a custom layout with keys assigned to Layer 3, including a small navigation layer (MO 3) and a symbols layer (MO 3).

Build a small navigation layer. On unused layer 3, select three positions and press Left, Down, and Right to assign them using Typing Binds a Key. Return to base and assign MO(3). The remaining layer-3 positions inherit through Transparent. Offline example.

Combine a tap key with a layer hold. Assign A, select it again, choose Layer Keys → LT, then choose layer 3. The resulting key is A on tap and layer 3 on hold. Offline example.

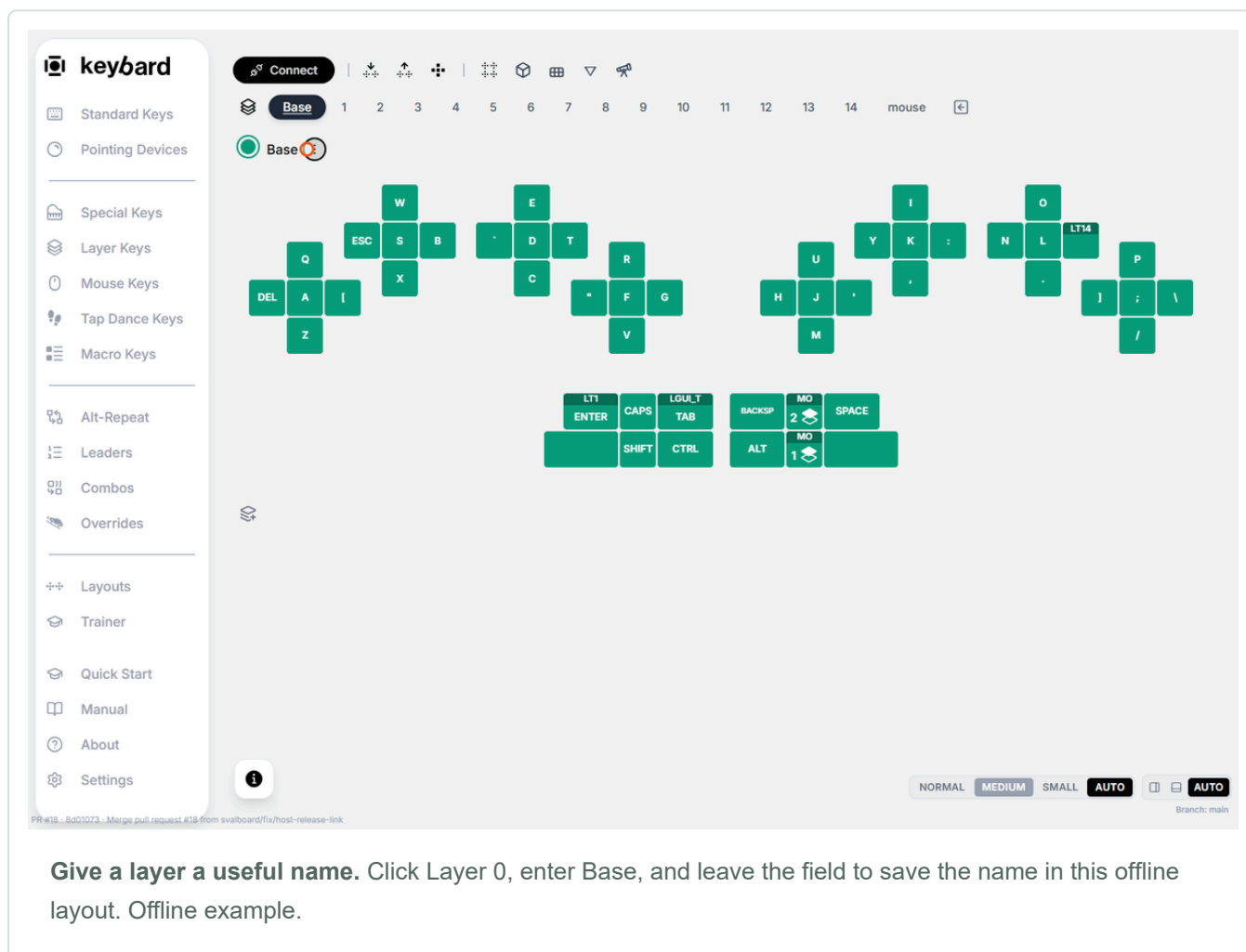
Build a simple navigation layer

1. Choose an unused layer, such as layer 3, and assign a small group of arrow keys from Standard Keys. Make the other positions transparent if they should inherit the base.
2. Return to the base layer and select a spare position for entering navigation.
3. Open **Layer Keys**, choose **Momentary (MO)**, and assign the tile for layer 3 to that position.
4. Review and apply the edits, then deselect the target. In a text editor, hold the new key and try the arrows. Release it and confirm normal typing returns.

For a tap action on the same key, assign its desired tap key first, select the position again if needed, then choose **Layer Tap (LT)** and its destination. The composer derives the tap action from the selected position.

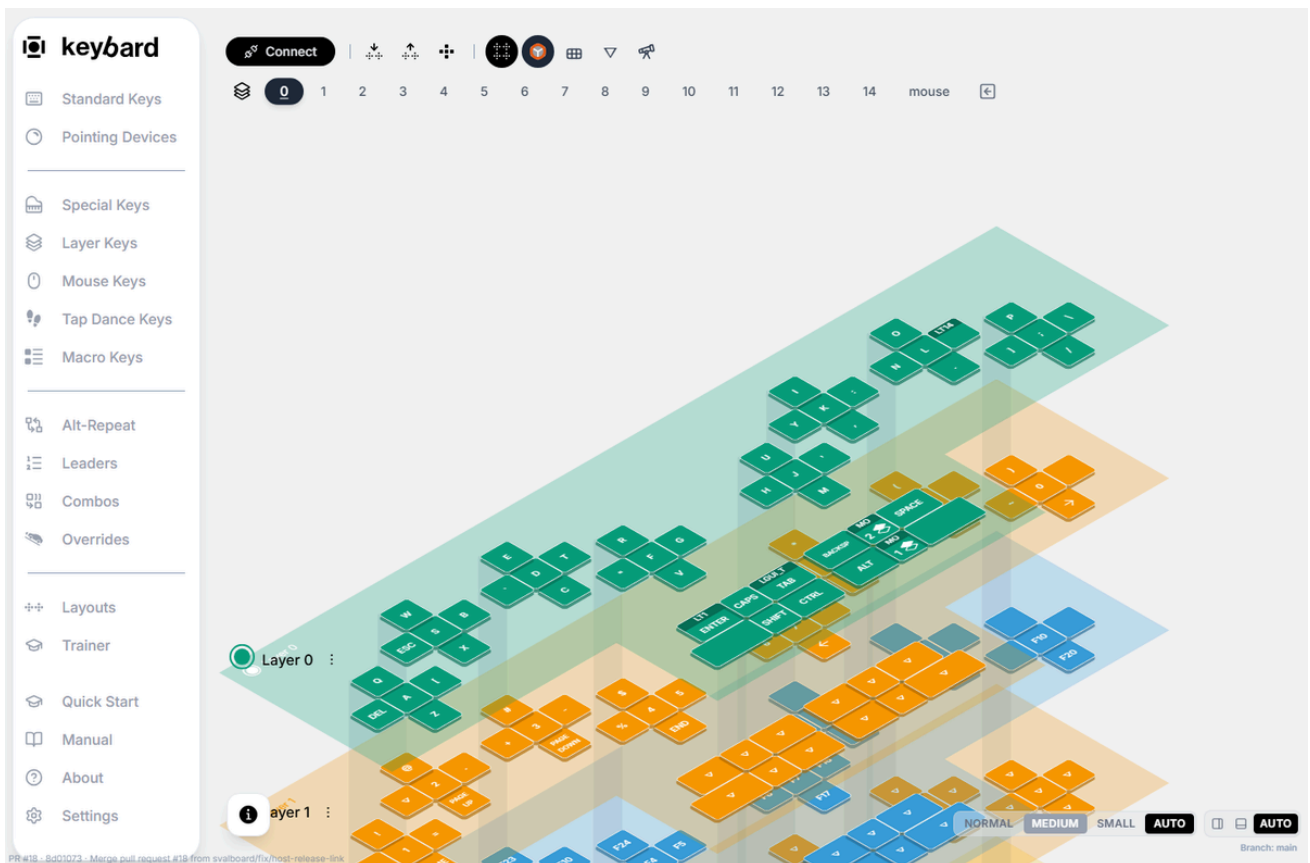
Layer Keys choice	When to use it
Momentary · MO	Activate a layer only while held.
Layer Tap · LT	Send a key on tap; activate a layer on hold.
Toggle Layer · TG	Turn a layer on or off with a tap. Keep a way to toggle it off.
Tap Toggle · TT	Hold momentarily, or tap the configured number of times to toggle.
One Shot Layer · OSL	Activate a layer for the next key.
To Layer · TO	Replace other active layers while retaining the default. Plan a return binding.
Default Layer · DF	Change the base for this power session.
Persistent Default Layer · PDF	Save the chosen base across power cycles.

Name, inspect, and reuse layers



The screenshot shows the Keyboard Layout Designer interface. On the left is a sidebar with a menu containing: keyboard, Standard Keys, Pointing Devices, Special Keys, Layer Keys, Mouse Keys, Tap Dance Keys, Macro Keys, Alt-Repeat, Leaders, Combos, Overrides, Layouts, Trainer, Quick Start, Manual, About, and Settings. The main area displays a keyboard layout with keys represented by green squares. The layout includes a top row with keys like W, E, I, O, and a bottom row with LT1, ENTER, CAPS, LGU,LT, TAB, BACKSP, MO, SPACE. A layer key labeled 'Base' is visible in the top left of the main area. At the bottom right, there are buttons for NORMAL, MEDIUM, SMALL, and AUTO, along with a branch selector set to 'main'.

Give a layer a useful name. Click Layer 0, enter Base, and leave the field to save the name in this offline layout. Offline example.



Compare layers together. Use Show Multiple Layers, then 3D View to change the presentation. Viewing a layer does not activate it on the keyboard. Offline example.

Save a reusable layer. Open the layer actions, choose Save Layer..., give it a name, and find it in Layouts. This is local browser storage, not a complete backup file. Offline example.

Click a layer name to rename it. Use its color control to make related layers easier to recognize. The layer's actions menu offers **Copy Layer**, **Paste Layer**, **Make All Blank**, **Make All Transparent**, and conversions between blank and transparent keys. Whole-layer operations affect many positions; check the target layer first.

Show Multiple Layers helps compare layers. **3D View** changes their presentation. **Hide All Transparent Keys** helps inspect inherited bindings; it does not rewrite those transparent keys. **Show All Layers** reveals layers hidden by the transparent-layer filter.

1. Open a layer's actions menu and choose **Save Layer...** to put a reusable copy in the local **Layouts** library.
2. Open **Layouts** and expand a saved or imported layout.
3. Drag the desired layer onto the destination layer in the editor.
4. Read the replacement confirmation and confirm only when the destination is correct.
5. Review and apply the resulting changes according to your update mode.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layers

Trainer

Quick Start

Manual

About

Settings

Layouts

Drag and drop to apply a layout to one of your layers or drag and drop individual keys. Default layouts are provided by Svalboard. You can save any of your own layers here, or import any .svil file.

Search layouts...

Import

sval-num-sym-layers

2 layers

0 Home-row numbers (default NavNAS)

1 Left-hand tenkey + right-hand nav

sval-default

4 layers

0 Layer 0

Connect

0 1 2 3 4 5 6 7 8 9 10 11 12 13

Layer 3

W E I O

ESC S B D T R U Y K : N L LT14 P

DEL A [X C * F G H J ' .

Z V M

LT1 ENTER CAPS LGU/LT TAB BACKSP MO 2 SPACE

SHIFT CTRL ALT MO 1

NORMAL MEDIUM SMALL AUTO

AUTO

Apply a layer from the library. Choose destination layer 3, drag the library's Layer 0 preview onto it, read the replacement confirmation, and choose OK. Check the destination before confirming. Offline example.

Start from a layout Svalboard provides

Layouts → sval-alt-alphas or sval-num-sym-layers

Besides your own saved layers, **Layouts** lists groups provided by Svalboard. Use any of their layers the same way as a saved one: drag it onto a layer in the editor.

Group	Layers	What they are
sval-default	The stock layers	The standard Svalboard layout, for getting back to a stock layer.
sval-alt-alphas	Dvorak, Colemak, HD Promethium, HD Neu, Graphite	Base layers for alternative letter layouts. Each is the stock base layer with only the letters and that layout's own punctuation moved, so your thumb keys and layer keys stay where they were. HD Promethium puts R on a left thumb key; Graphite adds Repeat on a right thumb key.
sval-num-sym-layers	Home-row numbers (default NavNAS), Left-hand tenkey + right-hand nav	Two numbers, symbols and navigation layers. The first is the factory NavNAS layer from the Svalboard firmware, with 1–0 on the center keys, shifted symbols above them and navigation below. The second puts 1–0 under the left hand as a tenkey, with arrows on the right home row. Their thumb keys are Transparent, so your base layer's thumb keys keep working.

keyboard

Standard Keys
Pointing Devices
Special Keys
Layer Keys
Mouse Keys
Tap Dance Keys
Macro Keys

Alt-Repeat
Leaders
Combos
Overrides

Layouts
Trainer
Quick Start
Manual
About
Settings

Layouts
Drag and drop to apply a layout to one of your layers or drag and drop individual keys. Default layouts are provided by Svalboard. You can save any of your own layers here, or import any .svil file.

Search: Dvorak Import

▼ sval-alt-alphas 1 layer

0 Dvorak

Layer 0

Connect | 0 1 2 3 4 5 6 7 8 9 10 11 12 13

Layer 0

DEL A \ ; ' ESC Q X ' E TAB LSFT P G [T] B N LT14 L
J Y U I D H F W V S Z
K M
LT1 ENTER CAPS LGUL TAB BACKSP MO 2 SPACE
SHIFT CTRL ALT MO 1

NORMAL MEDIUM SMALL AUTO AUTO

Branch: main

Put Dvorak on the base layer. With layer 0 selected, search Layouts for Dvorak, drag its preview onto Layer 0, and choose OK. The letters move; the thumb keys stay. Offline example.

1. Export a backup first if you are changing a layer you rely on, such as your base layer.
2. Select the destination layer in the editor. For a letter layout this is usually layer 0; for a numbers layer, the layer your NavNAS key opens (layer 1 in the stock layout).
3. Open **Layouts** and type a name, such as *Colemak* or *tenkey*, in the search box.
4. Drag the layer's preview onto the destination layer's name. Read the confirmation, which names both layers, and choose **OK**.
5. The destination keeps its own name but takes the layer's color; rename it if you like. Review and apply the changes according to your update mode.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layouts

Trainer

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Settings

Layouts

Drag and drop to apply a layout to one of your layers or drag and drop individual keys. Default layouts are provided by Svalboard. You can save any of your own layers here, or import any .svil file.

tenkey

Import

sval-num-sym-layers

1 layer

1 Left-hand tenkey + right-hand nav

Connect

0 1 2 3 4 5 6 7 8 9 10 11 12 13

Layer 1

NORMAL

MEDIUM

SMALL

AUTO

Branch: main

Try a different numbers layer. Select layer 1, search for tenkey, drag Left-hand tenkey + right-hand nav onto Layer 1, and choose OK. Offline example.

To try a letter layout without giving up your current one, drag it onto a spare layer instead, and add a **Default Layer** · **DF** key for that layer from Layer Keys. DF changes the base until the board restarts. You can also drag single keys out of a preview onto a position in your layout. Deleting a Svalboard group from Layouts is not permanent; Keyboard adds it back.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

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Settings

Standard Keys

English (US)

ESC F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12

1 2 3 4 5 6 7 8 9 0 - = BACKSP

TAB Q W E R T Y U I O P [\]

CAPS A S D F G H J K L ; ' ENTER

SHIFT Z X C V B N M , . / SHIFT

CTRL ALT GUI SPACE GUI ALT CTRL

Modifiers

None SHIFT CTRL ALT GUI

Mod, Mod-Tap, and One-Shot Mod

MOD MT MT QSM QSM

Blank and Transparent

Numpad

PRTS SCRLK PAUSE NUMLK = / *

INS HOME PGUP 7 8 9 -

DEL END PGDN 4 5 6 +

1 2 3 ENTER

← ↓ → 0 .

Function Keys

F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12 F13 F14

F15 F16 F17 F18 F19 F20 F21 F22 F23 F24

Connect Undo

1 2 3 4 5 6 7 8 9 10 11 12 13

Layer 0

W E I O

ESC S B ' D T R U Y K : N L LT14

DEL A [X C * F G H J ' P

Z V M

LT1 ENTER CAPS LGU/LT TAB BACKSP MO 2 SPACE

SHIFT CTRL ALT MO 1

NORMAL MEDIUM SMALL AUTO

AUTO

Branch: main

Choose Transparent or Blank in Keyboard. On layer 3, set the A position to Transparent, then Blank. Return to base: A is still there. The comparison below explains what happens when the upper layer is active. Offline example.

Local library, not a backup service. Saved layers and imported library items belong to this browser's local storage. Export important configurations to files; clearing browser data can remove local library content.

Transparent keeps the key underneath. Blank turns it off.

Suppose a position types **A** on your base layer. You hold a key to activate another layer. What should that same position do while you hold it?

Keep typing A

Set that position to **Transparent** on the upper layer.

Upper layer · active while held

▽ **Transparent**

Use the assignment underneath ↓

Base layer · underneath

A

Press this position → **A**

Make this position do nothing

Set that position to **Blank** on the upper layer.

Upper layer · active while held

Blank

Stop here. Do not use the key underneath.

Base layer · blocked

A

Press this position → **nothing**

Release the layer key: you return to the base layer, so this position types A again in both cases. Blank disables it only while the upper layer applies.

If you assign a different key—such as **1**—on the upper layer, the position types 1 while that layer is active.

CHAPTER 05 / BUILD USEFUL BEHAVIORS

Give a key more than one job

Special Keys groups additional actions such as media and system controls; **Layer Keys** and **Mouse Keys** have their own palettes. Use the same select-and-assign or drag-and-drop interaction as Standard Keys.

A normal assignment sends a key. A behavior can send a sequence, distinguish a tap from a hold, or recognize several keys together. Start with one small example, verify it in an empty document, and then build on it. The maintained Svalboard firmware provides 256 slots each for macros, tap dances, combos, overrides, alternate-repeat mappings, and leader sequences. Available entries depend on the firmware or file you have loaded.

Editing and using are different steps. A macro or tap dance needs both a definition and a key that invokes it. Combos and overrides respond to their configured triggers. Leaders need a Leader key as well as a sequence. In **Manual Updates**, apply the queue with **Apply N Changes** before testing on the board. In Live Updating, supported edits reach the board as you make them. Close the binding editor, deselect the assignment target, and focus a disposable text document before testing: typing while an assignment target is selected may change the layout.

Macros: send a short sequence

Macro Keys → **edit a slot**

A macro runs its actions in order. Begin with something harmless, such as the text “hello”. You can author a macro in an offline layout; executing it requires assigning it on a connected board.

keyboard

Standard Keys
Pointing Devices
Special Keys
Layer Keys
Mouse Keys
Tap Dance Keys
Macro Keys
Alt-Repeat
Leaders
Combos
Overrides
Layouts
Trainer
Quick Start
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About
Settings

Macro Keys

Send customizable sequences of keystrokes to trigger text strings, complex shortcuts, or automated actions with a single keypress.

Macro 0
0 "hello"

Macro 1
1

Macro 2
2

Macro 3
3

Macro 4
4

Macro 5
5

Macro 6
6

Macro 7
7

Macro 8
8

Macro 9
9

Macro 10
10

Macro 11
11

Macro 12
12

Macro 13
13

Macro 14
14

Macro 15
15

Macro 16
16

Connect | Undo | [Icons]

0 1 2 3 4 5 6 7 8 9 10 11 12 13

Layer 0

W E I O
ESC S B ' D T R U Y K : N L LT14 P
DEL A [X C * F G H J ' .
Z V M
LT1 ENTER CAPS LGU/L TAB BACKSP MO 2 SPACE
SHIFT CTRL ALT MO 1
NORMAL MEDIUM SMALL **AUTO** [Icons] **AUTO**
Branch: main

PR #18 - 8d07073 - Merge pull request #18 from...

Create and assign a text macro. Add a Text action containing hello, close the editor, and drag Macro 0 onto Q. Both the definition and the invoking key are now in the offline draft. Offline example.

1. Open **Macro Keys** and edit an unused numbered entry. In the compact bottom-panel presentation, the plus control opens an empty entry.
2. Add **Text** and enter `hello`. Give the macro a useful name by clicking its title where renaming is available.
3. Close the editor. Select a spare position in your layout, then choose that macro's numbered key tile, or drag its tile onto the position.
4. Apply queued changes if necessary. Deselect the assignment and focus an empty document. Press the physical key once; it should type your text.
5. Return to the editor to add actions only after the small example works. Export a fresh backup when you are happy with it.

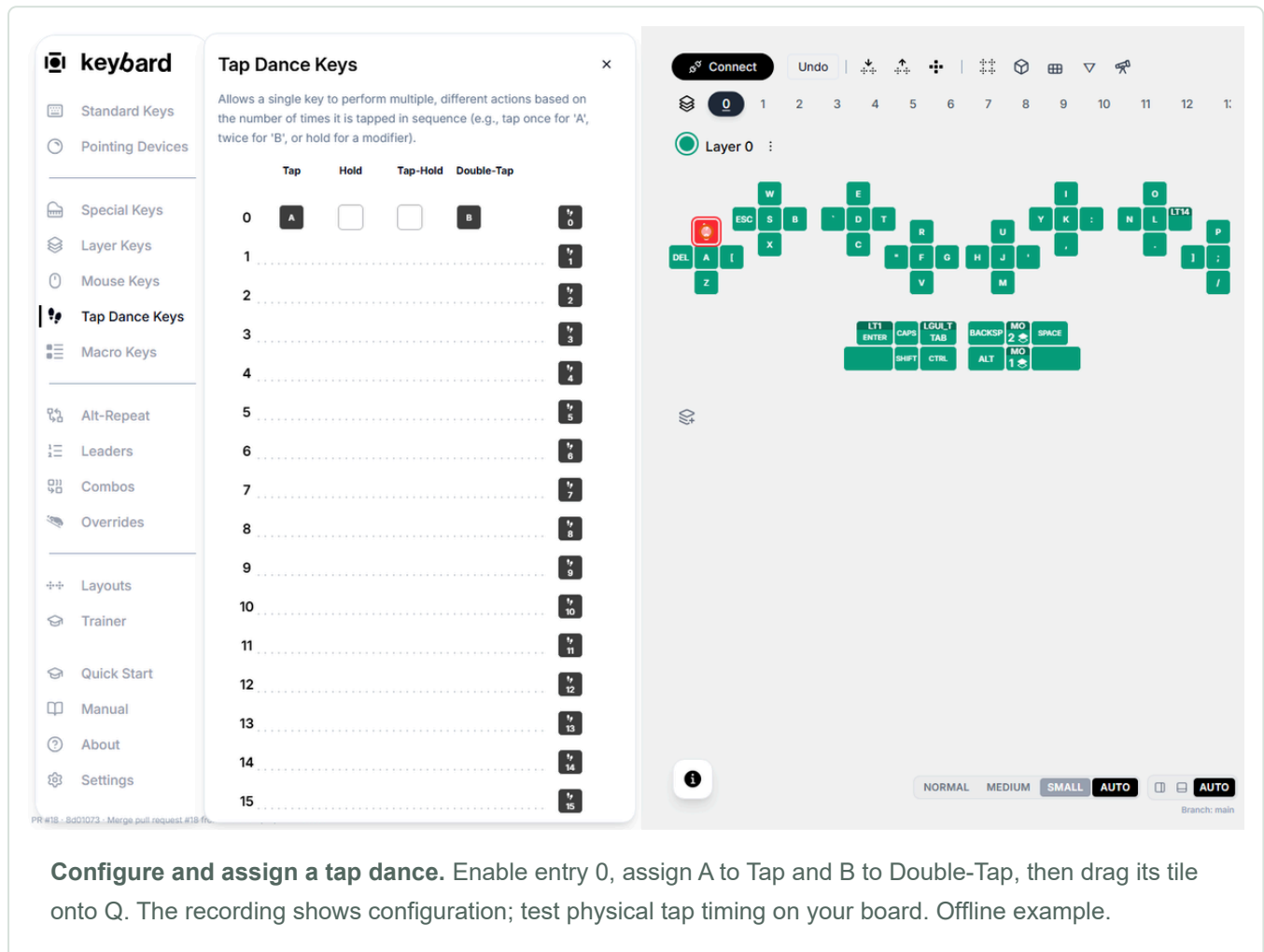
Key Tap presses and releases a key. **Key Down** holds it, and **Key Up** releases it. **Delay** inserts a pause in milliseconds. For a chord, use a modifier down, a key tap, and a matching modifier up. Always balance held keys with releases. Macro storage is shared across entries, rather than a separate fixed allowance for each macro.

Text macros send keyboard input, not clipboard contents. The operating system's keyboard layout affects the resulting characters; do not assume arbitrary Unicode text will work. Start with simple text and test punctuation in your actual OS layout. Avoid using passwords or other secrets as examples: a macro and an exported configuration are not a secure secret store.

Tap dances: tap, hold, or tap twice

Tap Dance Keys → edit a slot

A tap dance gives one physical position several outputs. A simple learning example is **Tap: A** and **Double-Tap: B**, with unused actions left empty. This makes the difference easy to see in a scratch document.



The screenshot shows the 'keybard' application interface. On the left is a sidebar with navigation options: Standard Keys, Pointing Devices, Special Keys, Layer Keys, Mouse Keys, Tap Dance Keys (selected), and Macro Keys. Below these are sections for Alt-Repeat, Leaders, Combos, Overrides, Layouts, Trainer, Quick Start, Manual, About, and Settings. The main area is titled 'Tap Dance Keys' and contains a table for configuring tap dances. The table has columns for Tap, Hold, Tap-Hold, and Double-Tap, and rows for entries 0 through 15. Entry 0 is configured with 'A' for Tap and 'B' for Double-Tap. To the right of the table is a keyboard layout diagram showing the placement of keys. The 'Q' key is highlighted with a red square, indicating it is the key assigned to the tap dance configuration. At the bottom of the keyboard layout, there are buttons for 'NORMAL', 'MEDIUM', 'SMALL', and 'AUTO', along with a 'Branch: main' indicator.

Configure and assign a tap dance. Enable entry 0, assign A to Tap and B to Double-Tap, then drag its tile onto Q. The recording shows configuration; test physical tap timing on your board. Offline example.

1. Open an unused entry in **Tap Dance Keys**.
2. Select the **Tap** slot and assign A. Select **Double-Tap** and assign B. **Hold** is a sustained press; **Tap-Hold** is a tap followed by a held press.
3. Review the timing field and leave **Enabled** on. Avoid changing timing and every action at once.
4. Close the editor and assign the entry's tap-dance tile to a spare keyboard position. Apply queued changes.
5. Deselect and test a single tap, then a quick double tap in an empty document. Adjust the timing if your deliberate double taps are being separated.

Release before editing. Release a held tap-dance key before changing its action. The launch firmware has a known edge case where editing a held dance can leave its previous output held.

Combos: press a small chord

[Combos](#) → [edit a slot](#)

A combo recognizes up to four input keycodes pressed within its timing window, and emits an **Output**. For example, A and B together can send Escape. These are keycode triggers, not a named pair of physical positions: consider where those assignments appear in your layers.

Make A + B send Escape. Enable a combo, choose A and B as its first two inputs, then set Output to Escape. No separate combo key is needed. Offline example.

1. Choose an unused combo. Fill the first two input slots with A and B and choose Escape for **Output**.
2. Leave the remaining input slots empty and enable the entry.
3. Apply changes. Deselect, focus a safe test window, and try each key separately before pressing them together.

4. If the chord is difficult to catch, review **Combo Timeout** in the Combos panel while connected. Increase it cautiously: a generous window may catch ordinary typing rolls.

Choose combinations that are comfortable but uncommon in your normal text. After verifying the mechanics, replace the demonstration with your preferred inputs. There is no separate assignment step beyond having those input keycodes available on the keyboard.

Modifiers, mod-taps, and one-shots

Standard Keys → Modifiers → Mod, Mod-Tap, and One-Shot Mod

A modifier chord, a held modifier, and a mod-tap are different tools. A chord such as Ctrl+C sends a shortcut. A mod-tap can send A when tapped and hold Control when held. A one-shot modifier lets you tap Shift, release it, and then type a shifted key without holding both keys together.

1. For the A/Control example, first assign A to the target position then select that position again if assignment cleared the selection.
2. In **Standard Keys**, select Control under **Modifiers**. The constructed keys use the currently selected position's base assignment.
3. Choose the generated **Mod-Tap** tile. Do not choose the standalone modifier or one-shot tile by mistake.
4. Apply changes, deselect, and test a tap in a scratch document. Then hold the key while pressing another key to verify the modifier behavior.

For a one-shot Shift, choose Shift and the generated one-shot tile instead. Test by tapping that key, releasing it, and typing a letter. Modifier controls include left/right choices; other composer presentations offer **MEH** (Control, Shift, Alt) and **HYPER** (those plus GUI). GUI is the Windows/Command-style modifier interpreted by your OS.

Use **QMK Settings** → **One Shot Keys** to review timeout and repeated-tap locking. Mod-tap decisions also depend on the Tap-Hold settings described below. Test typing rolls as well as isolated presses; a configuration that works slowly can feel different at speed.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

LAYOUTS

Trainer

Quick Start

Manual

About

Settings

Standard Keys

English (US)

ESC

F1

F2

F3

F4

F5

F6

F7

F8

F9

F10

F11

F12

CTL

1

2

3

4

5

6

7

8

9

0

BACKSP

CTL

TAB

Q

W

E

R

T

Y

U

I

O

P

[

]

CTL

CAPS

A

S

D

F

G

H

J

K

L

;

'

ENTER

CTL

SHIFT

Z

X

C

V

B

N

M

,

.

/

SHIFT

CTL

CTL

ALT

GUI

SPACE

GUI

ALT

CTL

Modifiers

None

SHIFT

CTRL

ALT

GUI

Mod, Mod-Tap, and One-Shot Mod

CTL

CTL

OSM

CTL

Blank and Transparent

Numpad

PRINT

CRCL

ANLSE

NUM

=

/

*

CTL

CTL

CTL

CTL

CTL

CTL

CTL

INSRT

NOME

PAGE

7

8

9

-

CTL

CTL

CTL

CTL

CTL

CTL

DEL

END

PAGE

4

5

6

+

CTL

CTL

CTL

CTL

CTL

CTL

UP

1

2

NUM

CTL

CTL

CTL

CTL

CTL

CTL

DOWN

0

.

CTL

CTL

CTL

CTL

CTL

CTL

Function Keys

F1

F2

F3

F4

F5

F6

F7

F8

F9

F10

F11

F12

F13

F14

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

F15

F16

F17

F18

F19

F20

F21

F22

F23

F24

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

CTL

Connect

Undo

1

2

3

4

5

6

7

8

9

10

11

12

13

Layer 0

CTL

ESC

W

B

E

D

T

R

U

Y

K

:

N

L

LT14

P

DEL

A

[

X

C

*

F

G

H

J

'

.

M

V

LT1

ENTER

CAPS

LGULT

BACKSP

MO

SPACE

2

SHIFT

CTRL

ALT

MO

1

1

NORMAL

MEDIUM

SMALL

AUTO

AUTO

Branch: main

Make one key A on tap and Control on hold. Assign A, select the position again, choose CTRL, then choose the generated mod-tap tile. Assignment can clear selection, so reselect before composing. Offline example.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layouts

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Settings

Standard Keys

English (US)

ESC F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

~ ! @ # \$ % ^ & * () - + BACKSP LSFT

TAB Q W E R T Y U I O P { } |

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

CAPS A S D F G H J K L ; ' " ENTER

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

SHIFT Z X C V B N M < > ? SHIFT

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

CTRL ALT GUI SPACE GUI ALT CTRL

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

Modifiers

None SHIFT CTRL ALT GUI

Mod, Mod-Tap, and One-Shot Mod

LSFT SFT OSM

Blank and Transparent

Blank Transparent

Numpad

PRINT CROLL ANLUSE NUM = / *

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

INSER HOME PAGE 7 8 9 -

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

DEL END PAGE 4 5 6 +

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

↑ 1 2 3 NUM

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

← ↓ → 0 -

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

Function Keys

F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12 F13 F14

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

F15 F16 F17 F18 F19 F20 F21 F22 F23 F24

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

Connect Undo

0 1 2 3 4 5 6 7 8 9 10 11 12 13

Layer 0

W E I O

ESC S B ' D T R U Y K : N L LT14

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

DEL A [X C * F G H J ' P

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

Z V M

LT1 ENTER CAPS LGULT TAB BACKSP MO 2 SPACE

LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT LSFT

SHIFT CTRL ALT MO 1

1

NORMAL MEDIUM SMALL AUTO

AUTO

Branch: main

Assign one-shot Shift. Select a position, choose SHIFT, and select the OSM tile. On your board, tap and release this key before typing the letter you want shifted. Offline example.

Overrides: change an action under a modifier

Overrides → edit a slot

An override replaces a trigger when its modifier and layer conditions match. A useful example is Shift+Backspace producing Delete. It changes how an existing assignment behaves, so you do not place an “override key” in your layout.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layouts

Trainer

Quick Start

Manual

About

Settings

Add Keys to Overrides

Standard Keys

English (US)

Modifiers

None SHIFT CTRL ALT

Mod, Mod-Tap, and One-Shot Mod

MOD MT OSM

Blank and Transparent

Numpad

Function Keys

Override 0

ON OFF

Trigger Replacement

BKSP → DEL

TRIGGER NEGATIVE SUSPENDED

Modifiers

NONE SHIFT CTRL ALT GUI

Layers

0 1 2 3 4 5 6 7 8 9 10 11

12 13 14 15

Active on trigger down ON OFF

Active on mod down ON OFF

Active on negative mod down ON OFF

Active trigger mod activates ON OFF

Reregister trigger on deactivate ON OFF

No unregister on other key down ON OFF

AUTO

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Make Shift + Backspace send Delete. Enable the entry, choose Backspace and Delete, and select Shift in both Trigger and Suspended. Suspending Shift means the replacement is Delete without Shift. Offline example.

1. Set **Trigger** to Backspace and **Replacement** to Delete.
2. In the **Trigger** modifier tab, select the Shift modifier you use. Review handedness if you use both Shift keys.
3. Use **Suspended** for the triggering modifier you do not want included with the replacement. **Negative** specifies modifiers that prevent the override; leave it empty for this basic example.
4. Select the intended **Layers**, enable the entry, and keep advanced activation options at their existing defaults unless you have a reason to change them.
5. Apply, deselect, and test against expendable text: Backspace should retain its normal action, while the configured Shift combination should perform Delete.

Leaders: type an ordered sequence

Leaders → Leader key and sequence entries

A leader sequence is ordered input after a dedicated Leader key. Unlike a combo, its keys are typed in succession. For a harmless first example, Leader, A, B can trigger a macro that types “hello”.

keyboard

- Standard Keys
- Pointing Devices
- Special Keys
- Layer Keys
- Mouse Keys
- Tap Dance Keys
- Macro Keys
- Alt-Repeat
- Leaders**
- Combos
- Overrides
- Layouts
- Trainer
- Quick Start
- Manual
- About
- Settings

Leader Sequences

Leader sequences trigger an output when you press a specific sequence of keys after the Leader key. Click on a key slot to assign a keycode.

LEADER

0 **A** → **B** → **Macro 0** ON OFF

1 ... →

2 ... →

3 ... →

4 ... →

5 ... →

6 ... →

7 ... →

8 ... →

9 ... →

10 ... →

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Connect Undo [Icons] 0 1 2 3 4 5 6 7 8 9 10 11 12 13

Layer 0 :

W E I O
ESC S B ' D T R U Y K : N L LT14 P
DEL A [X C * F G H J ' . I ;
Z V M

LT1 ENTER CAPS LGU/LT TAB BACKSP MO 2 SPACE
SHIFT CTRL ALT MO 1

1 [Info]

NORMAL MEDIUM SMALL **AUTO** [Icons] **AUTO**

Branch: main

Set up Leader, A, B. Enable the A-then-B sequence with Macro 0 as its output, then assign the Leader key. Create the hello macro in the preceding example first. Offline example.

1. Create and test the simple text macro first.
2. Open **Leaders**. Assign its **Leader** tile to a spare position.
3. Edit an unused sequence. Set its first two keys to A and B, and its **Output** to your macro. Enable it. Sequences allow up to five keys.
4. While connected, review **Leader Timeout**. **Per-key timing** resets that timeout after each key; otherwise the sequence shares its time allowance.
5. Apply, deselect, focus a scratch document, and press Leader followed by A then B within the allowance.

Alternate repeat: associate two actions

Alt-Repeat → edit a mapping

Alternate repeat uses the remembered key to choose an alternate action. For exploration, map A to B and assign the alternate-repeat action to a spare position. After typing A, invoke alternate repeat and inspect the result in a scratch document.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layouts

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Settings

Alt-Repeat Keys

Alt-Repeat keys remap what happens when you press Alt-Repeat after a specific key. Click on a key slot to assign a keycode.

ALT-REP

0 A → B

1

2

3

4

5

6

7

8

9

ON OFF

Connect

Undo

↓

↑

↕

↔

↻

⌂

▽

🔊

0 1 2 3 4 5 6 7 8 9 10 11 12 13

Layer 0

DEL

A

[

Z

ESC

S

B

X

W

E

D

T

C

R

F

G

V

U

H

J

M

Y

K

:

I

O

N

L

LT14

P

/

LT1

ENTER

CAPS

LGUL

TAB

BACKSP

MO 2

SPACE

SHIFT

CTRL

ALT

MO 1

1

NORMAL

MEDIUM

SMALL

AUTO

⏏

⏏

AUTO

Branch: main

Map A to B and assign alternate repeat. Enable the A-to-B mapping, then place the Alt-Rep key in the layout. This records configuration, not firmware execution. Offline example.

1. Choose an unused entry, set **Trigger** to A and **Alternate** to B, and enable it.
2. Assign the alternate-repeat action from the panel to a physical position. Apply the changes and deselect.
3. Test the simple mapping before exploring **Bidirectional**, **Ignore mod handedness**, or **Default to alternate**.

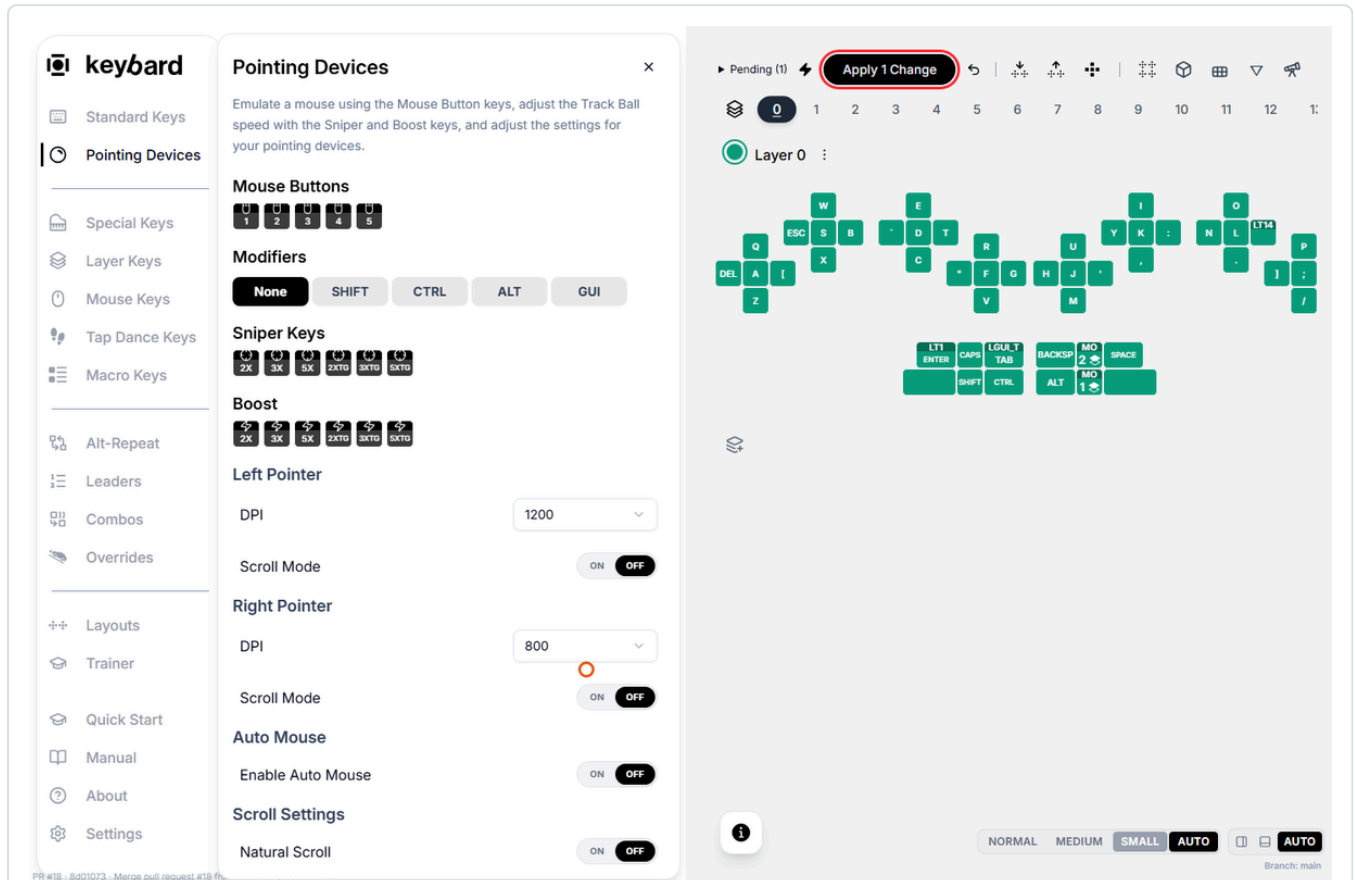
The launch firmware has known defects involving modifier matching and default-alternate behavior. Treat custom mappings as something to verify, especially with modifiers; do not depend on an untested mapping for a critical shortcut.

CHAPTER 06 / POINTING & MOUSE KEYS

Set up your pointing devices

Pointing Devices

Connect the board to see its sensor-specific controls. A loaded file can supply assignable pointing keys through the picker, but offline previews cannot demonstrate sensor behavior. The controls and ranges depend on the installed pointer, so your panel may differ from the example.



The screenshot shows the 'keyboard' software interface. On the left is a sidebar with navigation options: Standard Keys, Pointing Devices (selected), Special Keys, Layer Keys, Mouse Keys, Tap Dance Keys, and Macro Keys. Below these are sections for Alt-Repeat, Leaders, Combos, Overrides, Layouts, Trainer, Quick Start, Manual, About, and Settings. The main panel is titled 'Pointing Devices' and contains settings for emulating a mouse using keyboard buttons. It includes sections for Mouse Buttons (1-5), Modifiers (None, SHIFT, CTRL, ALT, GUI), Sniper Keys (2x, 3x, 5x, 2xTO, 3xTO, 5xTO), Boost (2x, 3x, 5x, 2xTO, 3xTO, 5xTO), Left Pointer (DPI: 1200, Scroll Mode: ON/OFF), Right Pointer (DPI: 800, Scroll Mode: ON/OFF), Auto Mouse (Enable Auto Mouse: ON/OFF), and Scroll Settings (Natural Scroll: ON/OFF). On the right is a keyboard layout preview for 'Layer 0' showing various key assignments. At the top of the preview is a 'Pending (1)' indicator and an 'Apply 1 Change' button. At the bottom are buttons for NORMAL, MEDIUM, SMALL, and AUTO, along with a branch indicator 'Branch: main'.

Adjust one pointer setting. Change the left pointer from 800 to 1200 DPI. This uses controlled test-board data to demonstrate the controls; it is not a recommended sensitivity or a sensor test.

1. Start with ordinary pointer movement. Adjust the left or right sensitivity in small steps, applying queued edits as needed.
2. Test scrolling and its direction. Review per-side scrolling controls, axis lock, and natural scrolling where available.
3. Assign a **Sniper** hold key to try slower, precise movement. **Boost** does the opposite. The 2×, 3×, and 5× versions choose the factor; toggle versions stay active without holding the key.
4. If you use automouse, confirm its target layer contains useful mouse bindings and a comfortable way back to ordinary typing. Check which pointer may activate it, then tune threshold, decay, and timeout one at a time.
5. Verify normal movement, scrolling, and layer return before exporting a backup.

Mouse Keys are keyboard-generated mouse movement and button actions; they are not the same as physical sensor sensitivity. TrackPoint recalibration applies to TrackPoint hardware. Avoid copying a sensitivity number between different sensor types and expecting identical movement.

Known firmware edge cases include buffered scrolling, combinations of speed factors, and disabling an active automouse layer. Keep the first configuration simple and verify the behavior you actually use.

CHAPTER 07 / SETTINGS & DIAGNOSTICS

Settings, timing, and board identity

Settings → General → QMK Settings...

The screenshot displays the QMK Settings application. On the left, a sidebar lists various settings categories: keyboard, Pointing Devices, Special Keys, Layer Keys, Mouse Keys, Tap Dance Keys, Macro Keys, Alt-Repeat, Leaders, Combos, Overrides, Layouts, Trainer, Quick Start, Manual, About, and Settings. The 'QMK Settings' panel is open, showing a list of settings including Magic, Grave Escape, Tap-Hold, Tapping Term (set to 210), Permissive Hold (new), Hold On Other Key Press, Retro Tapping (new), Quick Tap Term (set to 200), Tap Code Delay (set to 0), Tap Hold Caps Delay (set to 80), Tapping Toggle (set to 5), Chordal Hold, Flow Tap (set to 0), Auto Shift, Combo, One Shot Keys, and Mouse keys. A red box highlights the 'Apply 1 Change' button at the top of the settings panel. To the right, a keyboard layout diagram is visible, showing the current keymap configuration. At the bottom of the interface, a status bar indicates 'NORMAL MEDIUM SMALL AUTO' and 'AUTO'.

Stage a timing change. Change Tapping Term from 200 to 210 ms. Pending (1) and Apply 1 Change appear. Controlled test-board data; the recording does not write physical hardware.

Change appearance and turn off typing assignment. Try Dark, return to Light, and switch Typing Binds a Key off. These preferences belong to this browser. Offline example.

Settings has four sections: **General**, **Fragments**, **Backups** (see [Automatic backups](#)) and **Developer**. Browser choices include appearance, assignment behavior, and international palette selection. Choose a palette matching your OS keyboard layout; this does not change the OS layout itself. **Settings** → **Fragments** selects the finger and thumb assemblies used to draw your board. It describes the installed arrangement rather than rewiring it.

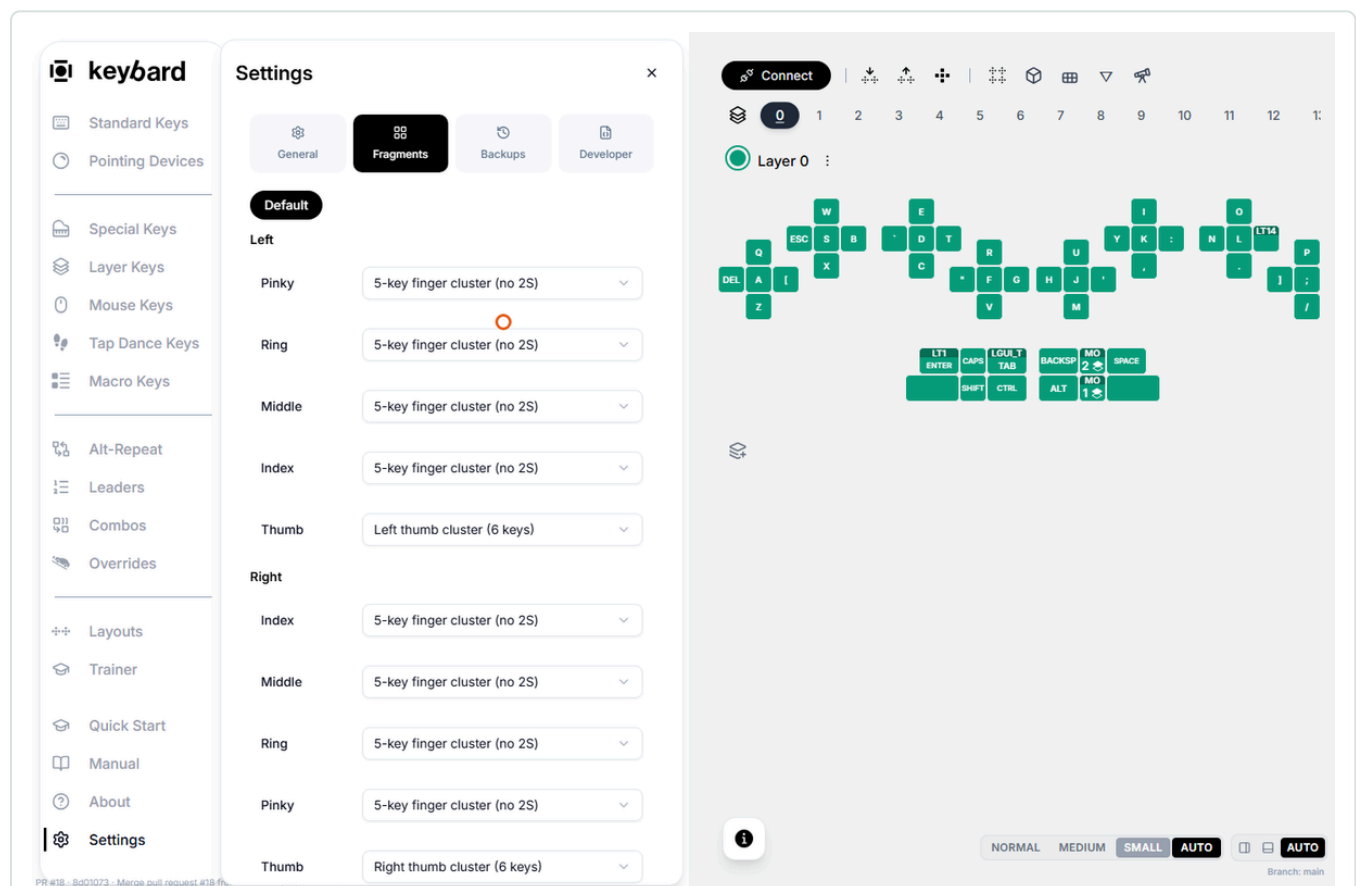
For timing, export a baseline first. Change one control, apply it, and test both slow deliberate presses and normal typing. **Tapping Term** influences tap-versus-hold decisions. **Quick Tap Term** affects repeated dual-role tapping. **Permissive Hold (new)**, **Hold On Other Key Press**, and **Retro Tapping (new)** control how other presses and releases affect that decision. **Chordal Hold** uses the hand relationship between keys; **Flow Tap** can favor taps during rapid typing and is off at zero.

Tap Code Delay controls generated tap duration, **Tap Hold Caps Delay** handles generated Caps Lock taps, and **Tapping Toggle** controls TT's tap count. Their launch defaults are 0 ms, 80 ms, and five taps; a stored toggle count of zero behaves as one. **Grave Escape** provides separate "Always send Escape" options for Alt, Control, GUI, and Shift.

A visible setting is not a guarantee that a firmware feature is enabled. Standard launch builds do not include Auto Shift, and some mouse-key settings have no runtime consumer. Use the launch notes for those limits rather than repeatedly adjusting an ineffective field.

Where supported, **Board name** changes how the keyboard appears to your computer. **Save now** writes immediately, even in manual mode; restart the keyboard for the USB name to change. Apply or export outstanding edits before restarting. Board identity is separate from a layout backup.

Settings → **Developer** → **Scan Lab...** is for advanced scanning, power, and firmware diagnostics. Its timing application and bootloader controls can affect hardware immediately. For a simple “does this switch register?” check, use Matrix Tester instead. Power estimates are not battery-life measurements, and longer idle intervals can delay the first response after inactivity.



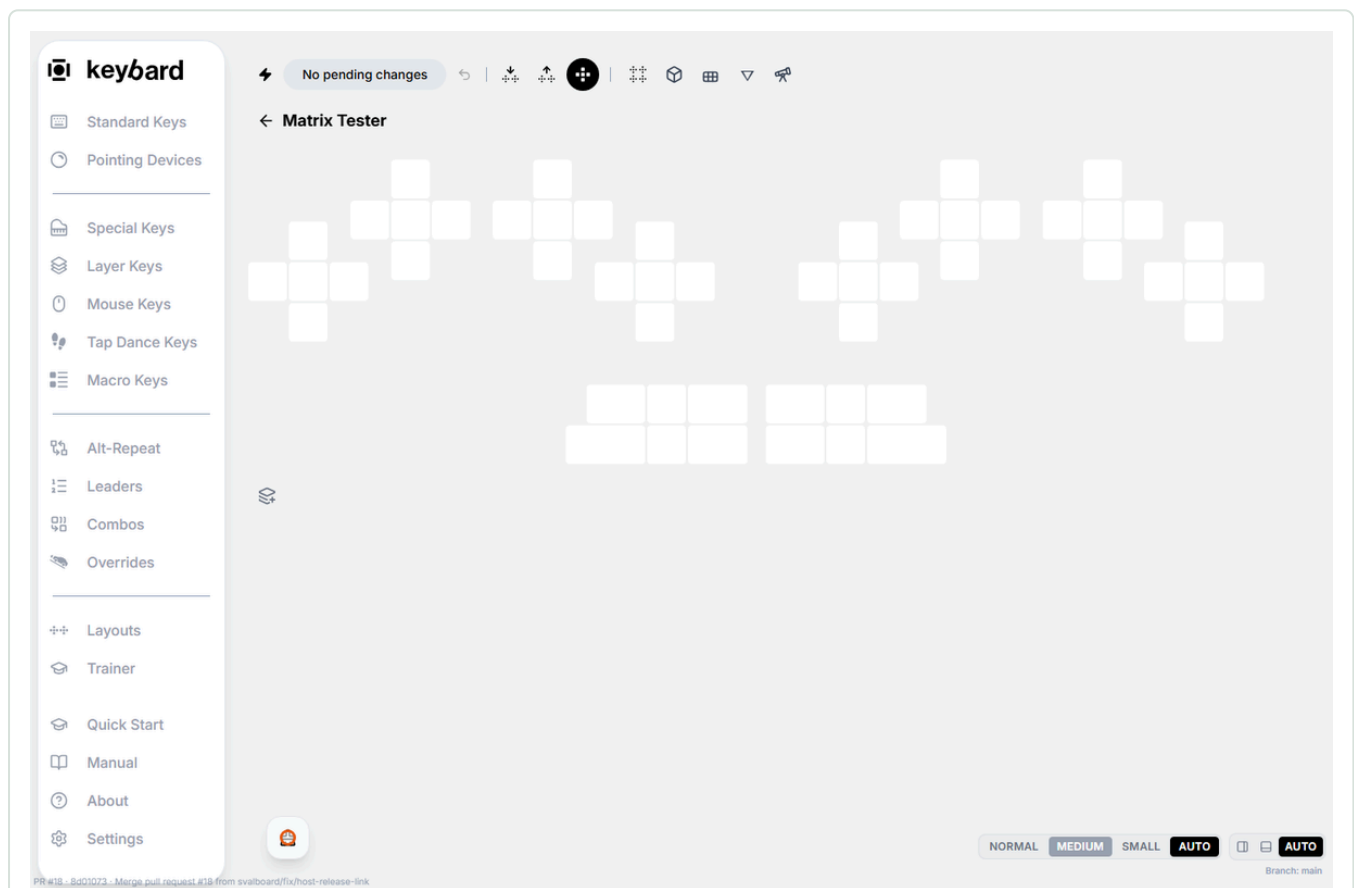
Change a fragment and restore it. Open Settings → Fragments, change the first finger cluster in the offline example, then restore its original selection. On your own board, choose the arrangement matching your hardware. Offline example.

Check every physical switch

Editor toolbar → Matrix Tester

1. Connect the board and open **Matrix Tester** from the toolbar.
2. Press each finger direction and each thumb key, holding briefly so the polling display can see it. Compare the highlighted position with the one you pressed.
3. Use **Clear Matrix** to clear the visual test history and try again. It does not erase your layout or board settings.
4. Exit Matrix Tester to return to editing.

This is a physical-switch check, not a monitor of the final text produced by macros, combos or tap dances. Very brief presses can fall between snapshots. If the matrix sees the switch but typing is wrong, inspect its layer and binding rather than assuming the switch failed.



Read and clear Matrix Tester history. Open Matrix Tester, see three switch reports, then use Clear Matrix. Reports are simulated test-board data; this is not a physical switch test.

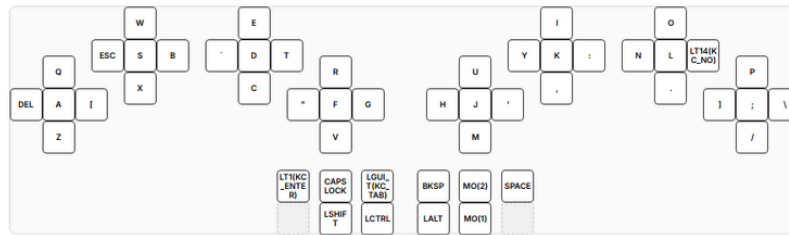
Print a desk reference

Open **Settings** → **General** → **Print Layers....** Review the non-empty layers and choose **Print**, then use the browser's print dialog to print them or save a PDF. A printed diagram is a learning reference; export a `.svil` as well for an editable backup.

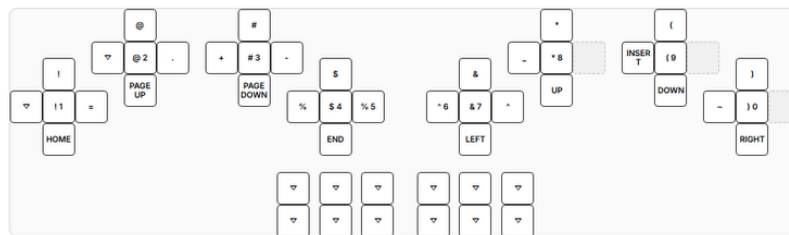
Unknown

4 layers with configured keys

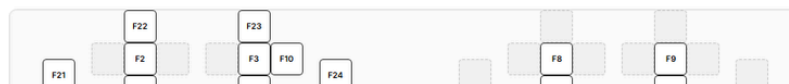
0 Layer 0



1 Layer 1



2 Layer 2



Prepare a printed layer reference. Open Settings → Print Layers..., review the layer count, then choose Print. The final frame shows Keyboard's actual print layout; the native print dialog is outside this browser recording. Offline example.

CHAPTER 08 / BACKUPS & REUSABLE LAYOUTS

Back up and import with confidence

Keyboard keeps two kinds of backup. **Automatic backups** are snapshots Keyboard saves on its own while a board is connected. **Exported files** are copies you make yourself and keep wherever you like. Use both: automatic backups catch the change you forgot to save, and an exported file survives a new computer or a cleared browser.

Automatic backups

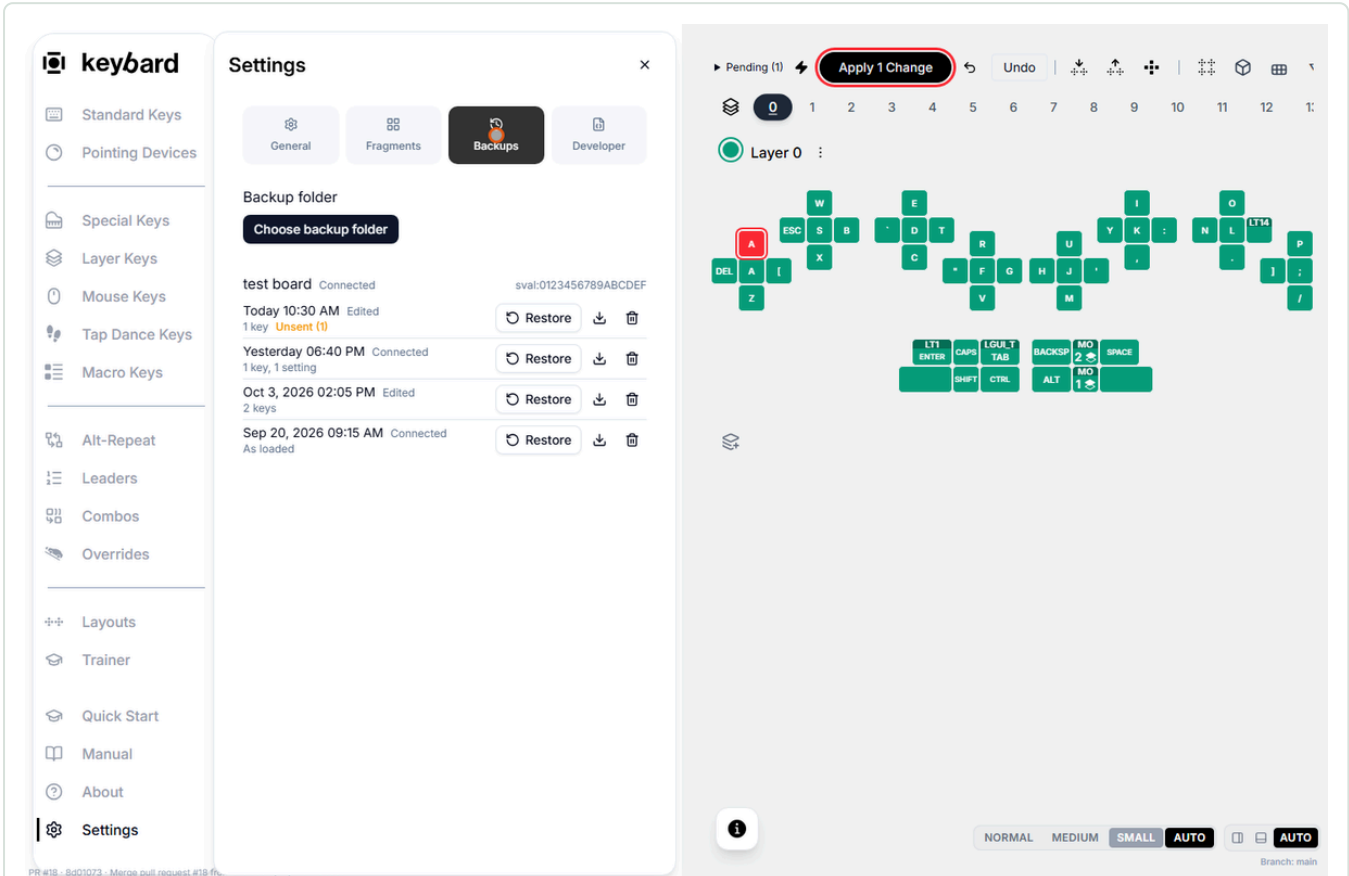
Settings → Backups

Nothing needs to be turned on. While a board is connected, Keyboard saves a snapshot of its configuration:

- when the board connects and finishes loading, before you change anything;

- about 10 seconds after your edits stop;
- when you switch to another tab or close Keyboard within those 10 seconds.

A snapshot holds the same information as an exported `.svil` file. If nothing changed since the previous snapshot, no new one is added.



A new backup appears after an edit. In Manual Updates, drag A onto Q. About ten seconds later, Settings → Backups lists a new Edited entry: 1 key, Unsent (1). Controlled test board; the earlier entries were added to show a history.

Backups are grouped by board, with the connected board first. Keyboard tells boards apart by their USB serial number (or, with older firmware, the keyboard's ID), so two Svalboards keep separate histories even if they share a name. Each entry shows:

Shown	Meaning
Date and time	When the snapshot was saved.
Connected or Edited	Saved when the board connected, or after you edited it.
Change summary	What changed since the previous backup of this board, such as <i>2 keys, 1 macro</i> . The first backup of a board says <i>As loaded</i> .
Unsent (N)	The snapshot includes N edits that were still pending in Manual Updates. Restoring it brings those edits back, even though the board never received them.

The newest eight backups of each board are shown; **Show all** lists the rest. Keybard keeps every backup from the last 24 hours, the latest backup of each day for 30 days, and after that the latest of each month. The most recent Connected backup is always kept.

Restore, download or delete a backup

- **Restore** opens the same **Review layout import** used for files. In Manual Updates, **Stage import** adds the differences to your pending changes; nothing reaches the board until you choose **Apply N Changes**. In Live Updating, **Apply import** writes them to the board straight away. Apply or discard pending edits before you restore.
- The download button saves the backup as a `.svil` file.
- The delete button removes that backup after you confirm.

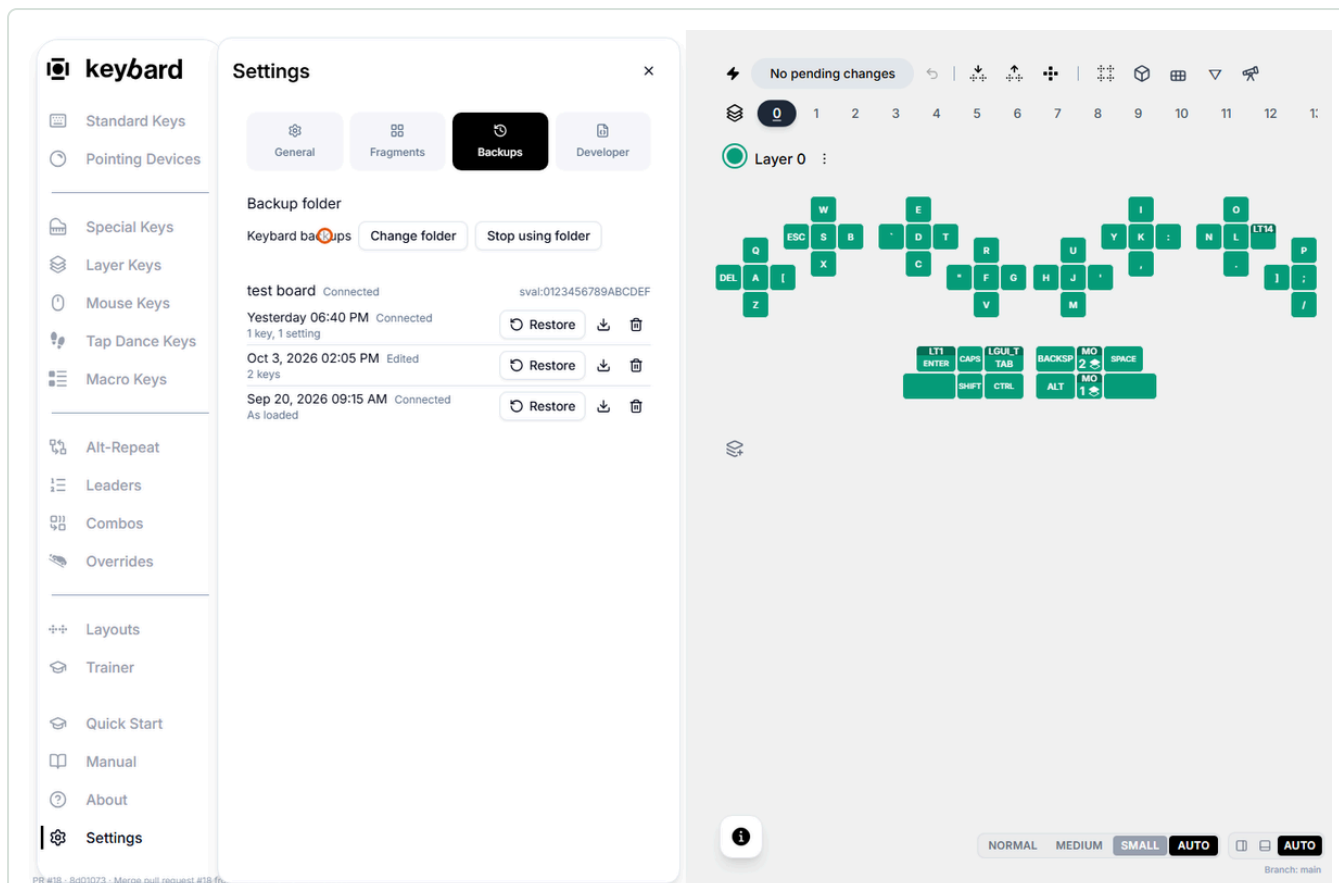
The screenshot shows the 'keybard' application interface. On the left is a sidebar with navigation options: Standard Keys, Pointing Devices, Special Keys, Layer Keys, Mouse Keys, Tap Dance Keys, Macro Keys, Alt-Repeat, Leaders, Combos, Overrides, Layouts, Trainer, Quick Start, Manual, About, and Settings. The main area is titled 'Settings' and contains a 'Backups' tab. Below this, it says 'Import staged. Use Apply to save it to the keyboard.' There are buttons for 'General', 'Fragments', 'Backups', and 'Developer'. Under 'Backup folder', there is a 'Choose backup folder' button. A list of backups is shown for a device named 'test board' (connected, sval:0123456789ABCDEF). The list includes: 'Yesterday 06:40 PM Connected' (1 key, 1 setting), 'Oct 3, 2026 02:05 PM Edited' (2 keys), and 'Sep 20, 2026 09:15 AM Connected' (As loaded). Each entry has a 'Restore' button. On the right, a keyboard layout is displayed for 'Layer 0'. At the top of the layout, there is a 'Pending (17)' indicator and an 'Apply 17 Changes' button. The layout shows various keys with their functions, including a custom key 'A' highlighted in red. At the bottom of the layout, there are buttons for 'NORMAL', 'MEDIUM', 'SMALL', 'AUTO', and 'AUTO'.

Restore an earlier backup. Choose Restore beside the October 3 backup, read the review, and choose Stage import. The differences wait under Pending until you apply them. Controlled test board; nothing is written.

Keep a copy in a folder

Automatic backups live inside the browser. To also keep them as ordinary files, choose **Choose backup folder** in Settings → Backups, pick a folder, and allow Keyboard to edit files there. This needs a browser that lets websites save to a folder, such as Chrome or Edge; elsewhere the button is unavailable.

- Each board gets its own subfolder, named after the board and its serial number.
- `latest.svil` always holds the newest backup. There is also one file per day with changes, named by date, such as `2026-10-03.svil`.
- Dated files are kept for 30 days, then one per month. Keyboard only deletes files it wrote itself; anything else you put in the folder is left alone.
- Renaming a board starts a new subfolder. The old one is left as it was.



Choose a backup folder. Choose backup folder, then Keybard shows the folder's name with Change folder and Stop using folder. It writes latest.svil and one dated file per day with backups. The recording replaces the system folder chooser with a stand-in folder.

Browsers ask again before a site may keep writing to a folder. When that happens, Settings → Backups shows **Paused** and a **Resume backups** button. Choose it, and if the browser offers **Allow on every visit**, choose that so later visits continue without asking. Backups made while paused are written when you resume. If the folder was moved or deleted, Keybard shows **Folder not found**; choose a folder again. **Change folder** switches to another folder, and **Stop using folder** stops writing files. Files already written stay where they are, and the browser's own backups continue.

Automatic backups have limits. They belong to this browser profile on this computer; another browser, profile or computer has its own list. Clearing this site's data deletes them, though copies in a backup folder remain. Only a connected board is backed up, not an offline example or file. Before a firmware update, export a `.svil` and keep it somewhere outside the browser.

Export and import files

Editor toolbar → [Export Layout or Import Layout](#)

Use native `.svil` for your working backups. The legacy `.vil` export retains QMK settings, but does not preserve all Sval-specific hardware values and metadata, including names, colors, and cluster selections.

Operation	What it preserves or changes
Export Layout	Downloads the editor's current configuration, including staged work.
Load File on welcome screen	Opens an offline draft; does not write a board.
Import Layout in editor toolbar	Reviews configuration changes for the current target.
Restore in Settings → Backups	Opens the same import review for an automatic backup.
Import in Layouts panel	Adds reusable layouts to the browser library.
Save Layer...	Saves one layer locally for reuse, not a complete device backup.

Import a compatible backup onto a connected keyboard

1. Connect the destination board and export its current configuration before making changes.
2. Apply or discard pending edits before beginning an import.
3. Choose **Import Layout**, select the file, and read **Review layout import**.
4. Check the destination, counts, warnings, and errors. **Back up current layout** is available in the review.
5. In Manual mode, choose **Stage import**, review pending changes, then **Apply**. In Live Updating, **Apply import** writes the changes. Offline, **Open layout** replaces the current draft.
6. Check the imported bindings and test important actions before relying on them.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layouts

Trainer

Quick Start

Manual

About

Settings

Layout opened offline. No keyboard was changed.

Connect

01234567891011121314mouse

Layer 0

QESCWB

DELAXIZ

W

ESCSBX

E

DT

C

R

F

G

V

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Y

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LT1

ENTER

CAPS

LGUIL_T

TAB

BACKSP

MO

SPACE

SHIFT

CTRL

ALT

MO

1

PR #18 - B401073 - Merge pull request #18 from svalboard/fix/host-release-link

Branch: main

Review and open an imported layout. Use Import Layout, read the offline review, then choose Open layout. On a connected board the review and write controls differ; see the steps below. Offline example.

Import is not a complete device-image restore. The current importer keeps the board's alt-repeat entries, leaders, one-shot configuration, layer colors, and fragment state. It reports unsupported fields and settings. Entries beyond the incoming file's layers and tables are kept rather than erased.

A file may be rejected for mismatched matrix dimensions, too many layers or table entries, incompatible keycodes, unsupported macro content, or capacity limits. Read the specific message instead of forcing the file through. Native backups preserve information even when a particular field cannot yet be restored automatically.

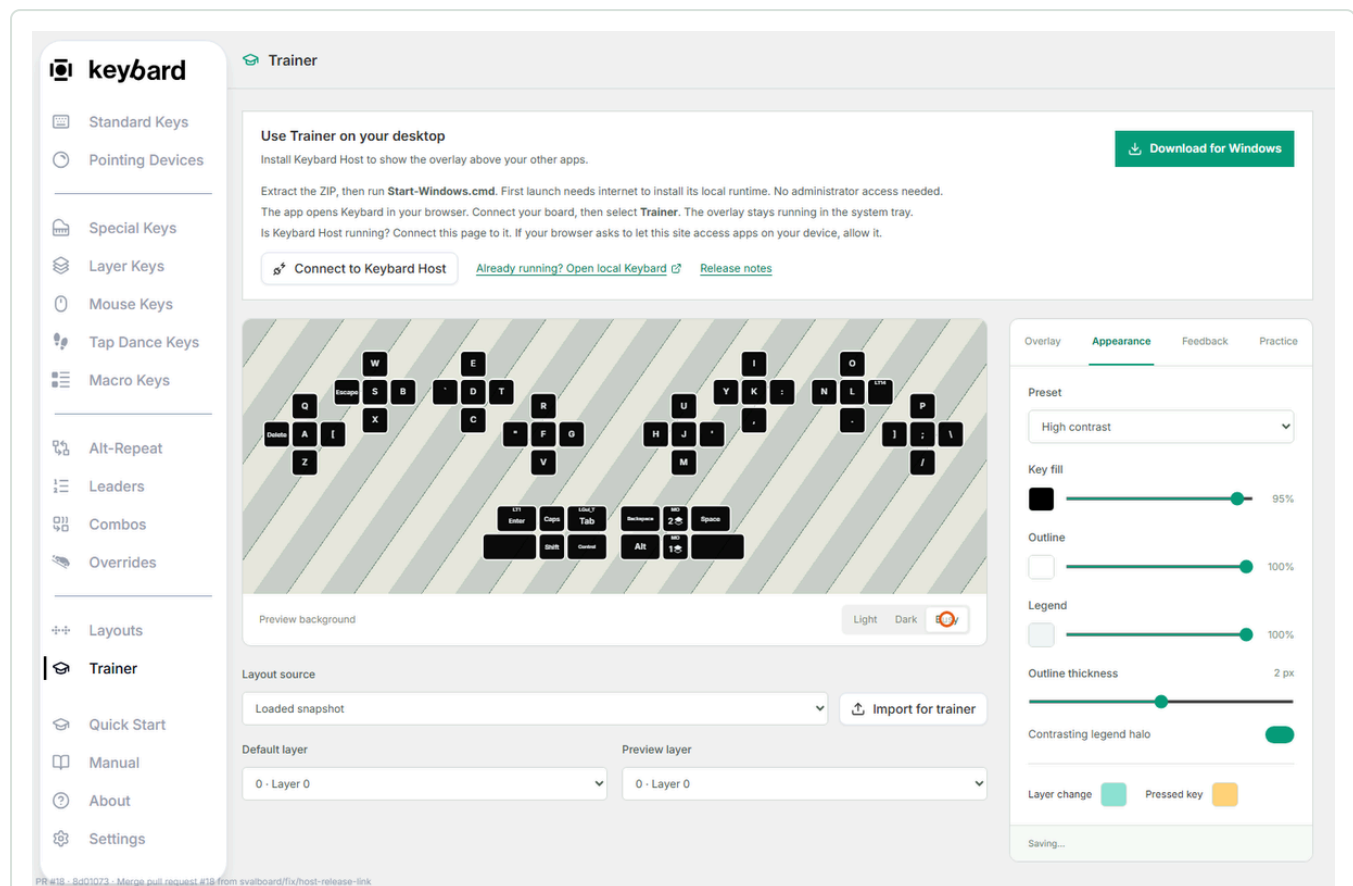
CHAPTER 09 / THE DESKTOP TRAINER

Trainer and the desktop overlay

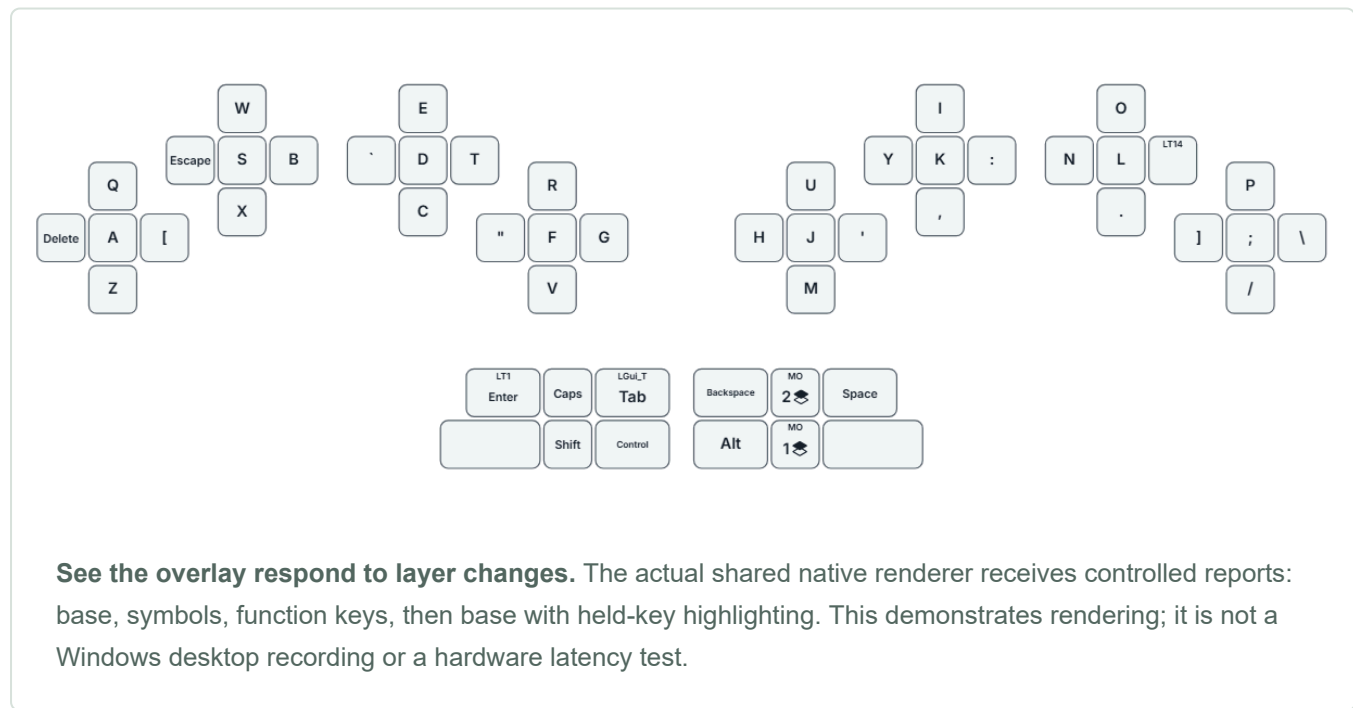
Trainer, directly below Layouts

Trainer previews the appearance and controls the optional desktop overlay. The browser is useful for choosing a style or reviewing an offline layout. Keyboard Host supplies the native

transparent window above your other applications and reads live board state directly. Its board operations are read-only.



Choose a readable overlay style. Apply High contrast, then try the Busy background to judge readability. This is the browser appearance preview.



See the overlay respond to layer changes. The actual shared native renderer receives controlled reports: base, symbols, function keys, then base with held-key highlighting. This demonstrates rendering; it is not a Windows desktop recording or a hardware latency test.

Install and show the Windows overlay

1. Select **Download for Windows** at the top of Trainer. Extract the ZIP and run **Start-Windows.cmd**.
2. Allow the first launch to download its private runtime. It needs internet, but no administrator rights, system Python, or PATH changes.
3. The app opens Keyboard. Connect your keyboard normally and select Trainer. If needed, use **Connect to Keyboard Host** and allow the browser's local-device access request. The install panel also links to local Keyboard when Host is running.
4. Choose the board in the host selector and use **Show overlay**. Host remembers its selected board for reconnection.
5. Drag the overlay into place. In **Overlay**, optionally enable **Click through keyboard**; the small handle remains available for moving or hiding it. **Place at bottom** provides a convenient starting position.

The companion stays in the system tray when you close the browser. Use **Hide overlay** or the overlay's handle controls to get it out of the way. After editing the keyboard layout, choose **Reload layout** so the companion refreshes its cached copy. If reconnection fails, check the host board selection and USB connection rather than importing a different layout to mask the problem.

The Windows download is a portable preview, not a signed installer. Linux source launch is available, with Wayland limitations for placement, stacking, click-through, and modifier reporting. macOS remains unverified; equivalent validated packages for all platforms are not promised.

Choose what you can see

Start with an **Appearance** preset, then adjust **Key fill**, **Outline**, and **Legend** colors and opacity independently. Try **Contrasting legend halo** over busy backgrounds. Use **Outline thickness** for definition without making the fill more opaque. **Hands** and **Overlay size** are in Overlay.

In **Feedback**, choose **Layer-change highlight**: Off, Quick flash, or Short fade, and its duration. Enable **Highlight held keys** for a matrix-tester-like indication. Matrix polling can miss very brief taps and does not describe every generated macro or combo output. The preview's **Preview held keys** button simulates a chord.

Current firmware reports active and default layers automatically, including changes during use. Older firmware may require **Default layer (older firmware)**. For offline previews, choose **Default layer** and **Preview layer**. **Layout source** distinguishes a loaded snapshot, editor draft, example, imported trainer layout, and a live read-only board. **Import for trainer** changes the preview source; it does not write that file to the keyboard.

- Standard Keys
- Pointing Devices

- Special Keys
- Layer Keys
- Mouse Keys
- Tap Dance Keys
- Macro Keys

- Alt-Repeat
- Leaders
- Combos
- Overrides

Layouts

Trainer

- Quick Start
- Manual
- About
- Settings

Use Trainer on your desktop

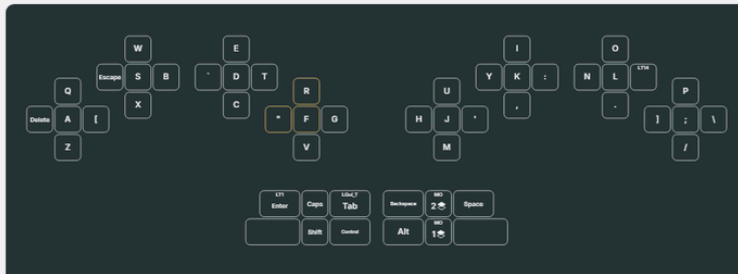
Install Keyboard Host to show the overlay above your other apps.

Extract the ZIP, then run **Start-Windows.cmd**. First launch needs internet to install its local runtime. No administrator access needed. The app opens Keyboard in your browser. Connect your board, then select **Trainer**. The overlay stays running in the system tray. Is Keyboard Host running? Connect this page to it. If your browser asks to let this site access apps on your device, allow it.

Connect to Keyboard Host

[Already running? Open local Keyboard](#) [Release notes](#)

Download for Windows



Preview background

Light Dark Busy

Layout source

Loaded snapshot

Import for trainer

Default layer

0 - Layer 0

Preview layer

0 - Layer 0

Overlay Appearance Feedback Practice

Layer-change highlight

Short fade

Duration 150 ms

Pressed-key color

Try a simulated chord. No key activity is being monitored.

Preview held keys

Saving...

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Preview highlighting. Choose Short fade and use Preview held keys to see a simulated chord. The browser preview does not measure native overlay latency.

- Standard Keys
- Pointing Devices

- Special Keys
- Layer Keys
- Mouse Keys
- Tap Dance Keys
- Macro Keys

- Alt-Repeat
- Leaders
- Combos
- Overrides

- Layouts

Trainer

- Quick Start
- Manual
- About
- Settings

Use Trainer on your desktop

Install Keyboard Host to show the overlay above your other apps.

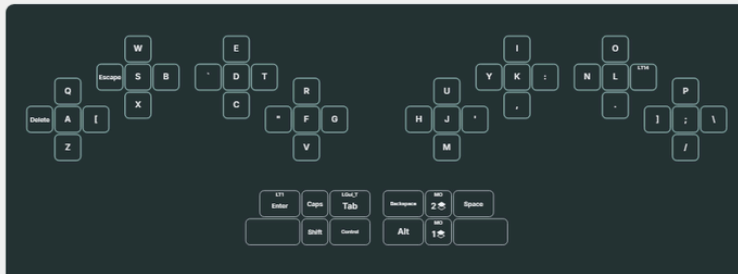
Extract the ZIP, then run **Start-Windows.cmd**. First launch needs internet to install its local runtime. No administrator access needed. The app opens Keyboard in your browser. Connect your board, then select **Trainer**. The overlay stays running in the system tray. Is Keyboard Host running? Connect this page to it. If your browser asks to let this site access apps on your device, allow it.

[Connect to Keyboard Host](#)

[Already running? Open local Keyboard](#)

[Release notes](#)

[Download for Windows](#)



Preview background

Light Dark Busy

Layout source

Loaded snapshot

[Import for trainer](#)

Default layer

0 - Layer 0

Preview layer

0 - Layer 0

Overlay Appearance Feedback Practice

Preset

Outline only

Key fill

0%

Outline

86%

Legend

100%

Outline thickness

1 px

Contrasting legend halo

Layer change Pressed key

Look at different layers in Trainer. Change Preview layer from base to symbols to function keys, then return to base. This only changes the preview.

Practice recall without recording your typing

The screenshot shows the 'keybard' web application interface. On the left is a sidebar menu with categories like 'Standard Keys', 'Pointing Devices', 'Special Keys', 'Layer Keys', 'Mouse Keys', 'Tap Dance Keys', 'Macro Keys', 'Alt-Repeat', 'Leaders', 'Combos', 'Overrides', 'Layouts', and 'Trainer'. The 'Trainer' section is active, showing a 'Quick Start' guide with instructions on how to use the trainer on a desktop, including downloading the app and connecting to the keyboard host. Below the guide is a large keyboard layout visualization. To the right of the keyboard layout are controls for 'Preview background' (Light, Dark, Busy) and 'Layout source' (Loaded snapshot, Import for trainer). Below these are 'Default layer' and 'Preview layer' dropdowns, both set to '0 - Layer 0'. On the far right is a 'Practice' panel with tabs for 'Overlay', 'Appearance', 'Feedback', and 'Practice'. The 'Practice' tab is active, showing a 'Recall practice' section with a toggle switch. Below this is a 'FIND THIS BINDING' section for 'LGui_T Tab' on the 'Left hand', with a 'Reveal' button and 'Remembered'/'Again' buttons. A 'Familiar bindings' section is also visible at the bottom.

Practice one recall prompt. Enable Recall practice, decide where the requested binding lives, Reveal it, then choose Remembered to advance. Grading is your own assessment.

1. Open **Practice** and enable **Recall practice**. Read the requested binding and hand, and decide where you would press.
2. Choose **Reveal**, then **Remembered** or **Again** to advance. This is not an automatic typing test.
3. For selective help, choose a key in the preview or the binding selector, click **Mark familiar**, and enable **Hide familiar legends**.
4. Turn practice off to return to ordinary reference. Closing the temporary controls also returns the desktop overlay to reference mode.

keyboard

Standard Keys

Pointing Devices

Special Keys

Layer Keys

Mouse Keys

Tap Dance Keys

Macro Keys

Alt-Repeat

Leaders

Combos

Overrides

Layouts

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Settings

Trainer

Use Trainer on your desktop

Install Keyboard Host to show the overlay above your other apps.

Extract the ZIP, then run `Start-Windows.cmd`. First launch needs internet to install its local runtime. No administrator access needed.

The app opens Keyboard in your browser. Connect your board, then select **Trainer**. The overlay stays running in the system tray.

Is Keyboard Host running? Connect this page to it. If your browser asks to let this site access apps on your device, allow it.

Connect to Keyboard Host

Already running? Open local Keyboard

Release notes

Preview background

Light Dark Busy

Layout source

Loaded snapshot

Import for trainer

Default layer

0 - Layer 0

Preview layer

0 - Layer 0

Overlay Appearance Feedback Practice

Recall practice

Self-assessed practice. Nothing is recorded from your typing. While following the board, recall also hides the desktop legends. Closing these controls returns the overlay to reference mode.

Familiar bindings

Select a key in the preview to mark its current binding familiar.

Left - Control - 0

Mark familiar

Clear

Hide familiar legends

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Hide a binding you already know. Select a familiar binding, choose Mark familiar, and enable Hide familiar legends. The preview hides that binding's legend.

Recall counts and familiar selections are temporary practice state, not a durable learning-history service. Appearance preferences are saved separately. When following the board, recall can also hide desktop legends; otherwise you are practicing against the selected preview layout.

CHAPTER 10 / RETURN TO YOUR WORK

Return to your work

After a refresh, Keyboard intentionally returns to the welcome screen. Previously permitted keyboards can appear under **or reconnect without the chooser**. Select the correct device to read its saved state again.

Return to a saved file. Choose Load File on the welcome screen and open a .svil file. The operating-system file chooser is not included in this browser recording. Offline example.

Connected boards are also backed up automatically. If you forgot to export, open **Settings** →

Backups: each board's recent sessions are listed there, in this browser. See [Automatic backups](#).

To continue offline work, choose **Load File** and reopen your exported draft. Browser preferences and the local Layouts library can persist, but they are not an automatic recovery copy of the current editor session.

Before each editing session, check the target, update mode, selected layer, and whether a key is selected. Use a small naming convention for backups: one before a major change, one after verification. Keep both until the new layout is comfortable.

CHAPTER 11 / TROUBLESHOOTING

Recover from common problems

I cannot connect

Check WebHID browser support, the USB data connection, the chosen device, and compatible firmware. Close other configuration applications if they are holding the device, then reconnect. A previously granted browser permission is not proof that the cable, firmware, or current device connection is working.

The layout changed, but typing did not

Check whether you are editing an offline draft or have pending Manual changes. Confirm the physical keyboard is on the layer you edited. Transparent keys may inherit a different binding, and a layer-tap key waits to distinguish a tap from a hold.

Typing unexpectedly changed a binding

Deselect the target and check **Typing Binds a Key**. In Manual mode, discard unwanted pending edits. If a change was already applied, explicitly restore the previous binding; discarding does not undo saved writes.

A save failed

Read the **Not saved** message. Pending edits remain, and some earlier writes may already have succeeded. If still connected, use **Retry**. If the device disconnected, **export the retained draft before reconnecting**. Reconnection reads the board again and may discard pending edits. An automatic backup saved after your edits also holds them, marked **Unsent**. Compare that state with your saved draft, then review an import if appropriate, observing the restore limitations above.

A name will not save

Layer, macro and tap-dance names have a 16-byte UTF-8 limit on supported firmware. Accented characters and emoji can use several bytes each. Shorten the name and read the error. A board's USB name is a separate setting with its own limits.

Controls or layers seem missing

Expand the navigation rail, reveal **Show editor controls** if the toolbar is collapsed, or use **Show All Layers**. Some settings depend on firmware capabilities and attached hardware; an offline example cannot demonstrate every connected-device function.

Keybard · Svalboard-QMK launch manual

Browser state, files and the board are different places to keep your work. Check the editing target, make one change at a time, and keep an exported copy you have verified.

[Back to the beginning ↑](#) · [Report an issue](#)